Chandhi Wijethilake

ETHICAL CHALLENGES IN UTILIZING ARTIFICIAL INTELLIGENCE IN BANKING OPERATIONS: A SYSTEMATIC REVIEW

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FACULTY OF INFORMATION TECHNOLOGY
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ABSTRACT

Wijethilake, Chandhi
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This research examines the ethical challenges associated with the adoption of Artificial Intelligence (AI) in banking operations through a systematic literature review (SLR). The review synthesizes findings from 72 peer-reviewed journal articles and conference papers published between 2018 - 2025, following a rigorous selection process based on PRISMA guidelines. The analysis reveals that, while AI technologies have transformed banking by enhancing operational efficiency, risk management, and personalized services, they also raise significant ethical concerns. Key ethical challenges identified according to the problematization include data privacy and security issues, algorithmic bias and discrimination, deficiencies in transparency, and accountability issues in AI-driven decision-making processes.

To comprehensively evaluate these challenges, the study employs a multi-theoretical framework that integrates ethical theories (utilitarianism and virtue ethics) to assess the normative implications, the Unified Theory of Acceptance and Use of Technology (UTAUT) to explore stakeholder perceptions and technology acceptance, and Actor-Network Theory (ANT) to elucidate the dynamic interrelations along with the translation among human and non-human actors in the banking ecosystem. Although the existing literature broadly acknowledges the existence of these ethical issues, most studies tend to discuss them superficially without detailed analysis or practical mitigation strategies.

Based on the synthesized evidence, this study proposes a novel conceptual framework promoting ethical AI adoption in banking. The framework highlights the interdependencies of ethical challenges among technical, ethical, and regulatory dimensions and provides a basis for future research aimed at developing more comprehensive ethical guidelines and governance standards. Ultimately, the study contributes valuable insights for both academia and practice, promoting responsible and ethically sound AI implementation in the financial sector.

Keywords: ethical challenges, artificial intelligence, banking

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1 INTRODUCTION

Humans have long been regarded as the primary embodiment of intelligence; however, with the advancement of technology and the evolution of artificial intelligence, this notion is increasingly being challenged. As AI systems continue to develop and exhibit cognitive capabilities traditionally associated with human intelligence, it becomes imperative to critically assess and redefine the concept of intelligence in the contemporary era. Artificial Intelligence (AI) can be named the most disruptive technology in recent history, fundamentally transforming various industries and everyday life. While the integration of AI in banking and finance has yielded considerable operational and strategic benefits, it has simultaneously introduced a range of ethical challenges. From a humanitarian standpoint, these ethical implications warrant rigorous academic attention. This study seeks to examine the ethical challenges related with AI adoption in the banking sector, as reflected in the current academic literature within a defined temporal scope. Through a systematic analysis and synthesis of secondary data sourced from reputable academic databases, the study aims to develop and propose a new conceptual approach to fostering ethical AI adoption in banking institutions.

Since AI's formal introduction in 1956 by the American computer scientist John McCarthy who initially coined the term, AI has evolved significantly, shaping modern technological advancements. McCarthy defined AI as *the science and engineering of making intelligent machines*, (McCarthy, 1959) a concept that has since expanded to encompass the automation of human cognitive processes through autonomous systems capable of problem-solving and decision making (Siemens et al., 2022) AI spans across a vast range of applications and disciplines. Since AI requires sophisticated computation, decreasing cost in hardware and dedicated AI chip design has made it more feasible to utilize. *The expansion of cloud-based services related to AI has also made it much more attainable for organisations would otherwise be hesitant (*Collins et al., 2021).

As a disruptive technology, AI has rapidly transformed the business and financial sectors. Its widespread adoption is driven by several key advantages, including automation of repetitive tasks, the ability to perform complex anal-

yses within seconds; significantly reducing computation time compared to human efforts and minimizing the impact of human error. Additionally, AI enhances efficiency by structuring processes in a more systematic and computational manner. Given these benefits, AI presents vast opportunities across various domains, making it an essential tool in modern industries. AI technology's widespread nature exerts a social influence since it has been impacting on daily lives as well as on communities all over the world, regardless any geographical limitations or cultural diversities.

Similarly, the utilization of AI in banking and financial context is also vital to scale and transform the processes to remain relevant with the current evolution of technology. The deployment of AI technologies can drastically improve the financial institutions' major outcomes such as higher profits, personalization, unique omnichannel experiences and innovation (Carson et al., 2020). The adoption of AI in banking has been increasing significantly, with financial institutions investing heavily in AI-driven solutions to enhance operational efficiency and service delivery. As of 2023, 75% of global banks have utilized AI-powered technologies in at least one operational area, AI has empowered banks to reduce operational costs by an average of 22%, while it has been estimated that AI has automated 30% of repetitive tasks in banking operations (Bhattacharya, 2024). These statistics highlight the rapid expansion of AI utilization in the financial sector.

However, alongside these advancements, concerns have emerged regarding the trustworthiness and ethical implications of AI models. High-profile cases of unethical AI behaviour across various sectors have prompted increased public scrutiny, leading AI developers to reconsider how ethical principles should be embedded in AI models (Belenguer, 2022; Hale, 2025; NBER, 2019; Vincent, 2019). The need for ethically robust AI solutions has become particularly critical in the banking and financial sector, which operates in a highly competitive and rapidly evolving technological landscape. The acceleration of AI-driven financial products and services necessitates a careful examination of the ethical dimensions of AI implementation.

Despite the practical necessity of extensively examining AI utilization in the banking sector, this research holds strong academic relevance by addressing critical gaps in the existing literature. While prior studies have acknowledged the ethical challenges associated with AI adoption across various sectors (Cath, 2018; Milossi et al., 2021; Manoj Kumar et al., 2023; Adeyelu et al., 2024; Chopra, 2024; Nguyen et al., 2024), they often lack a systematic and comprehensive analysis of how these challenges are categorized, evaluated, and addressed. Many studies focus on specific ethical concerns, such as bias or privacy, but few offer an overarching framework that synthesizes these challenges in a structured manner. Furthermore, much of the existing literature approaches ethical AI either from a technical or regulatory perspective, with limited research exploring the interplay and the intensified engagement between ethical considerations, organizational decision-making, and customer trust in the financial sector. This gap creates uncertainty regarding how banks can implement AI in a way

that is not only legally compliant but also ethically responsible and socially acceptable. This study aims to bridge this gap by integrating and synthesizing discussions on AI ethics in banking, categorizing key ethical challenges, assessing existing mitigation strategies, and proposing future research directions. The contribution of this research is academically significant as it provides a structured, theory-driven approach to understanding ethical AI adoption, thereby offering valuable insights for both researchers and practitioners in the development of responsible AI frameworks within the banking industry.

AI technology is continuously evolving, yet it has not fully replicated human cognitive abilities yet, particularly in areas such as empathy, moral reasoning, and ethical judgment (Vakkuri et al., 2020). The inherent complexity and interdisciplinary nature of ethics and human values make it challenging to integrate these principles into AI systems effectively. As ethical concerns surrounding AI deployment become increasingly critical, the development of fair and ethically responsible AI solutions has emerged as a key area of discussion within both academic and industry circles.

While AI presents significant opportunities for banking and finance, ethical concerns regarding the utilization of AI-enabled technology remain critical. This study conducts a Systematic Literature Review (SLR) to critically examine the potential ethical challenges, to categorize them based on the approach in addressing ethical landscapes, to assess the effectiveness of existing mitigation frameworks, future research directions and ultimately to propose a framework for responsible AI adoption in banking operations.

The structure of the thesis is organized to provide a coherent and systematic exploration of the ethical implications of AI adoption in the banking sector. The following section 2 details the theoretical foundation for the study, section 3 presents a preliminary background analysis to contextualize the study within the broader academic and industry discourse. Section 4 outlines the research methodology, detailing the systematic literature review (SLR) approach employed for data collection and analysis. Section 5 offers a synthesis of the findings derived from the SLR, followed by the discussion in section 6 including my extrapolation of results, which culminates in the development of a novel conceptual framework aimed at promoting ethical AI adoption in banking, based on insights from the secondary data. The thesis concludes with a key conclusion, and proposed directions for future research.

1.1 Research scope and objectives

The utilization of AI in the financial sector has been extensively examined in academic research over the past few years (Hentzen et al., 2022; Mohsen et al., 2024; Rahman et al., 2023). While existing studies acknowledge the potential risks and ethical implications of AI adoption (Qureshi et al., 2024; Ridzuan et al., 2024), they often lack comprehensive discussions on the concept of "ethicality" or the responsible use of AI. Additionally, specific ethical concerns related to

banking operations have not been sufficiently explored in academic literature (Doumpos et al., 2023; Lazo & Ebardo, 2023) and the regulatory efforts to address these challenges remain ambiguous. This gap in existing research highlights the need for a deeper investigation into ethical challenges woven around AI adoption in banking, and possible solutions to ensure the responsible deployment of AI in banking. Addressing these issues is crucial to fostering transparency, fairness, and trust in AI-driven financial systems. The research scope of the study spans as detailed below.

The study focuses on research published between 2018 and 2025, covering a seven-year temporal scope in which a rapid expansion of AI adoption was visible and leads to capture recent issues and emerging ethical challenges in this context. Conceptually, this study is aiming on identifying and analysing particular ethical concerns aroused by AI-driven banking operations. It is vital to employ a solid theoretical foundation to conduct this systematic exploration in an effective manner hence several theoretical concepts will be used as foundation lenses for proceeding the study. Ethical theories (Utilitarianism and Virtue Ethics) (Baumane-Vitolina et al., 2016) will be employed to evaluate the various ethical dimensions of the AI adoption in banking. Moreover, these theories provide a useful guidance to conceptualize the norm of ethicality, which is not found as a solid evident and allows to build a discussion on justification of accepted ethical behaviours in this domain. The Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2012) is used as a macro level lens to verify the integrity of performance expectations and social influence in organizational level in AI adoption. Actor-Network Theory (ANT) (Bencherki, 2017) is applied in a superficial manner to analyse AI utilization in banking as a socio-technical system, exploring the interactions between technology, financial institutions, regulatory bodies, and customers.

This study does not focus on a specific geographical region hence considers existing literature from a global perspective. A Systematic Literature Review (SLR) is adopted as the primary research methodology based on secondary data from the available academic literature. The results will be synthesized to provide a comprehensive categorization of the key ethical challenges in this domain and by examining the methodological approaches taken place in the secondary data to uncover the possible solutions or mitigations for the identified challenges. This synthesis will be beneficial to both academic and practical practitioners to continue future research and policy development as well. Since the concept of "ethical utilization" does not possess a solid definition due to its inherited context dependent nature, a broader discussion will be unfolded while analysing the literature gathered from different disciplines. Above objectives will be achieved by a critical examination of the literature and through an established theoretical framework based on the foundation theories depicted in next chapter.

2 THEORETICAL FOUNDATION

Establishing a robust theoretical foundation is essential to systematically analyse the ethical challenges accompanied with AI adoption in the banking sector. Given the interdisciplinary nature of this study, a comprehensive conceptual framework is necessary to integrate perspectives from technology, ethics, and financial institutions. AI, as non-human technology-based system, interacts with established social norms and ethical expectations, particularly in the highly regulated financial sector. This study aims to bridge the potential gap between AI technology adoption and its ethical implications by identifying the ethical interrelations and tensions among these three interconnected domains: technology, ethicality, and financial organizations.

2.1 Multi-theoretical approach

Due to the complex and often obscure nature of these relationships, a theoretical perspective is required to uncover underlying structures and dynamics. This study employs three distinct yet interrelated theoretical approaches: ethical theories; utilitarianism and virtue ethics, technology acceptance theory UTAUT, and socio-technical network theory; ANT. These perspectives collectively facilitate a structured analysis of AI utilization in banking, ensuring alignment with ethical principles while considering technological feasibility and institutional governance.

In this study, utilitarianism and virtue ethics illuminate the moral trade-offs and character-based responsibilities inherent in AI-driven banking decisions and operations; UTAUT highlights the organizational and social drivers of technology acceptance; and ANT reveals the socio-technical interactions among human and non-human actors. While these frameworks could be applied in far greater depth-drawing, for example, on comprehensive philosophical treatments of virtue ethics or advanced ANT studies of technology networks (Latour, 2005), this thesis deliberately employs only those facets that directly

illuminate the specific ethical challenges under investigation. By focusing on the elements that align most closely with the research questions, ensure that the theoretical foundation remains both targeted and practically applicable, without compromising the integrity or richness of the broader theoretical traditions.

2.1.1 Ethical theories: Utilitarianism and Virtue Ethics

Ethical theories provide a fundamental lens for evaluating the normative and moral implications of AI adoption in banking. The ethicality of AI-driven decisions and operations depends not only on legal compliance but also on broader moral justifications that shape societal perceptions of fairness, accountability, and trust. This study employs Utilitarianism and Virtue Ethics as discussed by Baumane-Vitolina, Cals & Sumilo (2016) as the primary ethical lenses to assess AI-related ethical concerns in banking.

"Utilitarianism, a consequentialist ethical theory, asserts that the morality of an action is determined by its overall contribution to the greatest good for the greatest number" (Bentham, 1994). In the context of AI adoption in banking, utilitarianism helps evaluate whether AI-driven decisions lead to optimal societal benefits, balancing efficiency gains against potential ethical risks such as bias, data privacy violations, or unfair lending practices.

AI adoption in banking is largely motivated by the pursuit of enhanced efficiency, cost reduction, and improved customer experience. According to utilitarian principles, the implementation of AI in areas such as credit scoring, fraud detection, customer service automation, and risk management should maximize benefits for both financial institutions and their clients. However, utilitarianism also raises ethical dilemmas when AI-driven decisions disproportionately disadvantage certain groups, such as marginalized communities by reinforcing existing biases in lending, financial risk assessment, or customer profiling according to a survey by National Bureau of Economic Research in 2019 (NBER, 2019). Therefore, this study critically examines the trade-offs between AI-driven benefits and its unintended ethical consequences.

Virtue Ethics, attributed to Aristotle, focuses on moral character rather than the consequences of actions (Baumane-Vitolina et al., 2016). It emphasizes the development of virtues such as fairness, honesty, integrity, and responsibility in decision making. Applying this theory to AI in banking highlights the necessity for ethical AI governance, where financial institutions and AI developers are expected to embody virtuous behaviours in designing and implementing AI systems. Virtue ethics are particularly relevant in addressing accountability and transparency in AI decision-making. Unlike utilitarian approaches that assess ethicality based on outcomes, virtue ethics emphasizes the moral responsibility of AI developers, financial institutions, and regulators in ensuring AI-driven decisions align with ethical values. For instance, if an AI-based credit scoring system results in unfair loan denials, a virtue ethics perspective would demand accountability from the institutions deploying the

technology, urging them to uphold fairness and integrity in AI development and application.

By integrating utilitarianism and virtue ethics, this study develops a dual ethical lens to examine both the practical benefits of AI in banking and the moral obligations of institutions to ensure AI is used responsibly.

2.1.2 Unified Theory of Acceptance and Use of Technology (UTAUT)

The adoption of AI in banking is also a matter of technological acceptance and implementation at the organizational level. The Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003) is a widely recognized framework for examining factors that influence technology acceptance in the field of Information Systems. There are four major constructs, performance expectancy, effort expectancy, social influence and facilitating conditions in the model. This study applies UTAUT as a superficial lens to analyse how AI adoption in banking is shaped by organizational expectations, performance concerns, and ethical considerations.

Specifically consider the construct *performance expectancy* in above framework for this study which refers to the perceived benefits and efficiency gains of adopting AI in banking. According to Lazo & Ebardo, (2023), banks implement AI mostly to improve credit risk assessment, fraud detection, and automated customer service, with expectations that these technologies will enhance financial decision-making in efficient and reliable manner. However, performance-driven AI adoption must be balanced with ethical considerations such as bias mitigation, transparency, and fairness. As an instance, while AI-driven credit scoring models may improve efficiency in loan approvals, they risk perpetuating discrimination if trained on biased historical data. UTAUT helps explain why banks prioritize AI adoption for operational improvements while potentially overlooking ethical concerns, highlighting a critical tension that this study seeks to explore.

Considering the construct *effort efficiency*, it refers the level of ease or perceived usefulness related with the using of a particular system or technology. In this thesis, effort expectancy from UTAUT is employed to analyse how the perceived ease of use of AI technologies in banking influences user acceptance across different stakeholders. It helps explain how ethical challenges such as lack of transparency or complexity of AI systems may increase perceived effort, thereby negatively affecting adoption. This construct links ethical concerns with user trust, accessibility, and willingness to engage with AI-driven banking solutions.

Another core construct of above framework is *social influence* which refers to the degree to which individuals perceive that important other (e.g., peers, supervisors, institutions) believe they should use a particular technology. Ethical AI adoption in banking is highly dependent on public trust and regulatory compliance. If customers view AI based decision making as unfair or biased, banks face reputational risks and resistance to AI-driven financial services

(Adamyk et al., 2024; Lazo & Ebardo, 2023). By integrating UTAUT, this study examines the relationship between AI ethical concerns and its acceptance within banking institutions, considering how ethical dilemmas such as lack of transparency, bias, and data privacy risks or affects stakeholder perceptions and influence AI adoption decisions.

2.1.3 Actor-Network Theory (ANT)

The ANT theory which has been originated around 1970 and has been widely expanding in many disciplines is employed in the study with relevance to both human and non-human actors (Bencherki, 2017; Callon, 1986; Callon & Law, 1997) in organizational context to examine AI in banking as a socio-technical system. ANT conceptualizes both human and non-human entities: people, technologies, regulations, even data as "actors" whose interactions form and transform dynamic networks.

Latour (2005) argues that to grasp how networks form and evolve, one must "follow the actors" through their negotiations, since the translation process or how messages are interpreted, reshaped, and re-articulated; fundamentally determines what travels across a network. It is vital to differentiate between intermediaries and mediators which have been manipulated in this study later. An *intermediary* carries meaning or force from one point to another without alteration, acting as a passive conduit whereas a *mediator*, by contrast, transforms the message it carries introducing unpredictability as actors reinterpret, reframe, or repurpose information within the network. This concept will be utilized in depth to discuss my framework later in this study.

Callon's (1986) four moments of translation; problematization, interessement, enrolment, and mobilization, are also been used to reveal how these networks co-construct what counts as knowledge, authority, or ethical practice. The four moments can be defined according to Callon as: *Problematization*: The phase in which a focal actor defines a problem and identifies other actors as indispensable allies in addressing it. *Interessement*: The process of deploying devices or strategies to lock other actors into the roles outlined during problematization. *Enrolment*: The negotiation and agreement among actors to accept, modify, or reject the roles assigned to them. *Mobilization*: The stage where spokespersons or representatives act on behalf of a broader network, solidifying collective engagement and action. This concept in ANT developed by Callon, is significant in depicting the process of translation of ethical use of AI in the banking ecosystem which is also discussed in-depth later in the study.

AI can be considered as a socio-technical construct in the perspective of ANT which merely seemed to be a computational tool. Unlike frameworks that treat AI as an isolated technology, ANT considers AI as part of a network of human and non-human actors, including financial institutes, AI developers and regulatory institutions, customers or end users, and data and algorithmic models. This framework allows in-depth exploration of dynamic interactions and behaviours while adopting AI in banking context. Due to the widespread use of

AI in current landscape, it is embedded within social, economic and legal structures inevitably. ANT's theoretical lens allows to study and analyse how AI dynamically interacts with other actors not only humans but also regulatory frameworks like GDPR and customer trust dynamics, how the translation of ethicality among the varied networks would impact on holistic comprehension of ethical challenges in AI deployment in banks and financial institutions.

ANT can be used to highlight the distributed nature of responsibility in AI adoption. The ethical issues such as bias in credit approvals, privacy risks in data disclosure are not solely the responsibility of AI developers but also banks, regulators and policy developers are also included parties in it (Rahman et al., 2023; Waller et al., 2025). This viewpoint is also crucial in understanding the complexity of determining ethical principles and best practices since different stakeholders influence in varied manners to shape AI adoption and its results.

The integration of ethical theories, UTAUT and ANT frameworks, a robust theoretical foundation has been established for the study to analyse the ethical dimensions of AI adoption in banking operations. This multi-theoretical approach ensures that the ethical challenges of AI adoption are evaluated not only from a moral perspective but also considering the organizational and social impacts on it. Utilization of multiple theoretical lenses also avoids author's bias on analysis and provides a holistic framework for comprehensive study.

2.2 Research questions

According to the existing knowledge in the domain, AI has been widely adopted while transforming the core banking operations to digitalization along with enhanced efficiency and customer satisfaction. But the ethical implications of the potential AI utilization are often underexplored and vaguely discussed in the existing literature.

The problematization of this research was challenging, as the concept of "ethicality" is inherently context-dependent and lacks a universal definition. Furthermore, the tensions between the opportunities presented by AI and the potential ethical concerns highlight the need for a deeper examination of this issue.

If legal frameworks were established to define ethical and non-ethical behaviour, AI developers could leverage these regulations to train models in a universally accepted manner. Yet, such universal framework could not be obtained since the concept of ethicality varies depending on many contexts related to human behaviour and even cultural and geographical foundations also. However, ethical principles such as fairness, equality, autonomy, veracity, confidentiality, and fidelity (Melé, 2024) are inherently complex and often lack clear-cut guidelines for implementation and evaluation. The absence of standardized measures to manage or quantify these ethical dimensions presents a significant challenge, making it difficult to navigate and enforce ethical AI de-

velopment effectively. To address these complexities, the following research questions have been formulated to guide this study.

RQ1 – What are the ethical challenges associated with AI utilization in banking operations?
RQ2 – What are the key ethical challenges categories identified/discussed in the existing literature?
RQ3 - What methodological approaches have been used to analyse and evaluate the ethical implications of utilizing AI in banking operations?

The first two research questions aim to explore the established knowledge in the current research landscape, highlighting the potential ethical implications of AI adoption. The theoretical background for this RQs is based on utilitarianism and virtue ethics, which can be utilized in a broader perspective to examine the ethical use of AI. Moreover, UTAUT framework provides an ample theoretical lens to understand the technology adaptation in terms of performance expectancy, effort expectancy, social influence and facilitating conditions. Accordingly, the use of AI in banking enhances both performance expectancy and effort expectancy, which in return leads to improved behavioural intention while adopting technology. For the analysis of the existing literature, ANT will be referenced in a superficial yet a broader perspective to identify and understand the socio-technical interactions and interdependencies among various actors in this banking ecosystem, how the translations occur in different networks to transmit the concepts of ethical AI adoption. The referenced theoretical foundations would be a solid basis to conduct the research in an academically relevance manner. The findings from the existing literature will be categorized based on their nature and theoretical foundation, helping to identify recurring ethical challenges and their rationale.

Additionally, analysing the methodological approaches used by early researchers will provide insight into existing gaps and limitations. Understanding these methods will be essential for assessing how ethical AI challenges in banking have been studied and where further research is needed.

3 ETHICAL LANDSCAPE OF AI ADOPTION

Artificial Intelligence has become an integral part of various sectors, transforming industries into complex sophisticated landscapes. As an umbrella term encompassing a range of machine-driven, non-human tasks, AI continues to evolve rapidly, integrating seamlessly into different domains. Considering banking and financial context, while its adoption offers significant advantages, it also raises ethical concerns, particularly in banking operations, where issues such as bias, privacy, and accountability come into question. This chapter explores existing research on AI adoption in banks, the ethical considerations surrounding its use, and the theoretical foundations that guide this process. By examining these areas, this section aims to provide a comprehensive understanding of the ethical challenges associated with AI in banking in the current literature. This section serves as a foundational prelude to the study, establishing the necessary background for the final synthesis of key findings derived from the SLR, which will be presented in detail in section 4 and then discussed in depth later chapters.

3.1 AI in current banking and financial landscape

Traditional banking relied on manual intervention for critical operations such as credit scoring, fraud detection, account management, risk management, and customer service (Galazova & Magomaeva, 2019). However, these processes were heavily dependent on extensive documentation, resulting in time-consuming analysis and decision-making while requiring skilled labour. Consequently, the reliance on manual processes reduced operational efficiency and negatively impacted financial outcomes due to excessive time and labour costs. The increasing adoption of AI-powered solutions has revolutionized the way financial services are delivered, enhancing operational efficiency, risk management, and customer experience (Mossavar-Rahmani & Zohuri, 2023). AI is the outcome of a collection of tools created to replicate human data collection,

learning, and decision-making abilities. These tools have the capacity to learn and function independently, but they can also be used in an assistance capacity. As a result, there are now different categories of AI applications that could either automate or enhance human work (French & Shim, 2024).

According to an analysis conducted by Vakkuri, Kemell, Kultanen and Abrahmsson in 2020, almost half of the industries were utilizing AI in the software development process which is a positive impact on the AI models to have a responsible approach in ethical aspect of AI. Various AI driven applications are now shaping different areas within organizations from voice assistants and biometric systems in front office operations for customer interaction, to fraud detection, credit assessment, and compliance processes in internal tasks, and way to back-office functions where credit underwriting supported by smart contracts (Fares et al., 2023). Machine Learning (ML) algorithms, which form an integral part of AI, enable financial institutions to offer personalized services by leveraging predictive analytics and automated decision-making. Financial institutions have increasingly integrated AI-driven technologies to enhance fraud detection, credit risk assessment, and regulatory compliance (Jáuregui-Velarde et al., 2024). AI-powered chatbots and virtual assistants are widely used in the banking sector to increase personalized customer experience. An empirical study on how the virtual assistants enhance empathy and customer satisfaction, results of the study show that chatbots and virtual assistants improve customer satisfaction and service quality in the banking industry (Pfoertsch & Sulaj, 2023). The study emphasized that chatbots which can handle customer engagement along with empathy significantly enhance customer experience and are considered a critical component of service excellence.

AI-adoption in banks also contributed to addressing the challenges with human errors which impact on quality of service and time taken to provide services. The use of AI has enhanced the service quality, error free service and rendering efficient services (Mogaji & Nguyen, 2022). This can be emphasized as a major motivation to adopt AI in banking operations to eliminate human errors and possible bias to facilitate the customers with more enhanced, reliable, and trustworthy services.

Hence, the banks manage vast amounts of highly sensitive personal data obtained by KYC (Know-Your-Customer), which should only be used for purposes explicitly consented to by customers and not beyond the initially authorized scope (Lazo & Ebardo, 2023). The extensive use of customer data in AI-powered banking operations has raised ethical concerns regarding responsible technology use. The large-scale utilization of banking data for developing and training AI solutions must be conducted ethically, ensuring that customer intentions and privacy are respected. Machine learning (ML) models rely on trained data, which may contain human errors, potentially leading to biases, privacy risks, and other ethical challenges (Bhatt & Meniya, 2022).

While AI has revolutionized banking operations, its ethical implications remain a growing concern. Several real-world controversies highlight the risks associated with unethical AI implementation in banking and finance. One of the

most controversial cases is Apple Card's credit limit discrimination case (Vincent, 2019). Customers complained that even in cases where their financial backgrounds were comparable, women were given substantially lower credit limits by Apple's AI-driven credit rating system, which was run by Goldman Sachs. This sparked widespread criticism and regulatory scrutiny, raising concerns about transparency and fairness in AI-driven credit assessments highlight the need for supervision on ethical aspects of AI driven technology. Similarly, cases found in-relation to discrimination in lending practices by AI driven models in banking where AI biased caused 80% of black mortgage applicants to be rejected (Hale, 2025) and where minority black and Hispanic borrowers were charged higher interest rates compared to white borrowers (NBER, 2019).

3.2 The conceptualization of "ethical use" of AI

"Ethics" is concerned when discussing moral standards of a social context or a community. Ethics in economical rationality as; *"Ethics combined with logic, epistemology and metaphysics is a traditional cognitive sphere that originated in Ancient Greece"* (Baumane-Vitolina et al., 2016). Accordingly, it may be regulatory or descriptive. A particular social group's existing norms and behavioural standards are studied by descriptive ethics, which also aims to comprehend the processes involved in ethical reasoning and decision-making. Regulatory ethics, on the other hand, uses logic and basic moral principles to theoretically examine moral standards. Considering these theories in a technological context of adopting AI in a financial institute, once human intervention has been augmented or automated by a machine or an agent (French & Shim, 2024), the behavioural context covered by the human intervention must be also considered in order to fill the gap properly. In this adoption, AI can be used to automate or assist the banking operations according to the requirement and nature of the tasks.

Existing research acknowledges the ethical issues and challenges of utilizing AI in different contexts including banking and finance. But often demonstrates a lack of systematic analysis of the scope, in relation to the ethical dimensions in depth. The major concern is that the norm of "ethical use" does not inherit a clear boundary hence it is massively dependent on the contextual application (Mudrack & Mason, 2013). The rationale of "ethicality" is self-evident in the context it is embedded in (Tenbrunsel et al., 2010).

Ethical principles vary significantly across individual beliefs, cultural contexts, and geographical boundaries. For instance, in many Asian societies, ethical considerations are deeply intertwined with religious and philosophical traditions, influencing daily life and decision-making. In contrast, Western cultures often emphasize pragmatic benefits and individual autonomy over collective ethical concerns (Peters & Carman, 2024). This divergence highlights the complexity of defining "ethical behaviour" in a universally accepted manner. In the context of AI and machine learning, this complexity poses a significant challenge in developing algorithms that can accurately and robustly interpret ethi-

cal principles. Since ethical behaviour is inherently humanistic and context-dependent, encoding it into AI systems requires nuanced approaches that account for cultural and societal differences (Chernysh et al., 2024). The absence of a universally accepted ethical framework further complicates AI training, making it difficult to establish clear guidelines for what is considered ethical or unethical across different communities worldwide (Kurshan et al., 2022).

Throughout history, human behaviour has evolved alongside technological advancements. The discovery of fire led to transformative changes in early human lifestyles, while the invention of the wheel marked a significant milestone in human progress. Similarly, disruptive technologies have consistently reshaped human civilization. Today, as we navigate an era dominated by artificial intelligence (AI), questions arise regarding the potential for AI to surpass human evolution. With the development of self-learning algorithms and complex networking systems, AI has the capacity to influence and redefine human evolution in unprecedented ways (Ridzuan et al., 2024).

The widespread adoption of AI raises critical ethical concerns, particularly in relation to human perceptions of what is considered right or acceptable versus wrong or unacceptable. Ethical theories, which have evolved over centuries, continue to play a fundamental role in societal decision-making, as humans remain central to ethical discourse. Theories such as Utilitarianism and Virtue Ethics provide valuable frameworks for assessing the ethical implications of AI adoption, ensuring alignment with established humanistic principles. According to Baumane-Vitolina, Cals and Sumilo (2016) ethical theories are traditionally categorized into three major groups: teleological, deontological, and virtue ethics. These frameworks provide insight into the potential positive and negative consequences of actions. Utilitarianism, a widely recognized teleological theory, has been extensively applied in economics to explain rational decision-making. It assumes that efficiency and the pursuit of greater satisfaction underpin human morality. As articulated by Bentham, *"The greatest happiness of the greatest number is the foundation of morality and legislation"* (Bentham, 1994). This perspective emphasizes that ethical actions should consider the well-being of all affected parties. Hence, utilization of particular technology should not negatively impact upon the well-being of affected parties.

In contrast to Bentham, Baumane-Vitolina, Cals and Sumilo (2016) argue that deontological theories advocate for the adherence to universal ethical principles, regardless of specific outcomes. These theories highlight the significance of moral duty and the underlying intentions behind actions, rather than merely focusing on consequences. Deontological ethics assert that ethical behaviour is a fundamental obligation rather than means to an end, reinforcing the necessity of ethical principles as guiding standards in decision-making. Virtue ethics on top of the above two types emphasizes that taking ethically correct decisions and presence of traits such as honesty, sense of responsibility and fairness, that impact on goal achievement of certain social group or organization in a broader context. The significance of adherence to ethical principles and regulations while adopting technology is timely important in the humanitarian perspective.

AI, being an umbrella term for various technologies as depicted in figure 1, has been broadly embedded in almost all the technologies that have been used to enhance the operational efficiency and accuracy of banking operations. AI technologies such as Machine Learning (ML) and Natural Language Processing (NLP) has been commonly embedded in banking core systems to facilitate more efficient predicting purposes and automated customer advisory (Bhatt & Meniya, 2022; Eskandarany, 2024; Jáuregui-Velarde et al., 2024; Kumar et al., 2021; Lagasio et al., 2022; Polireddi, 2024). Hence, there is a broader and in-depth discussion to be carried out to examine the impact of ethical concerns on the utilization of AI in banking.

AI as an umbrella term for many tools and technologies:

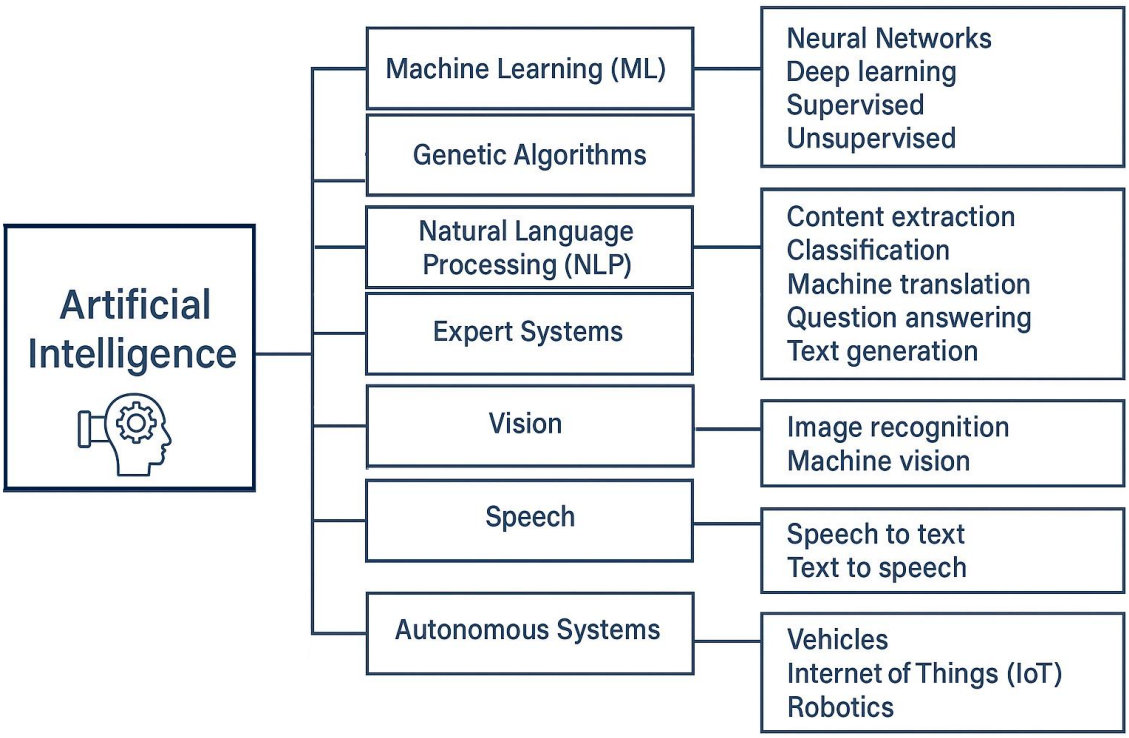


FIGURE 1 Types of AI (French & Shim, 2024)

Considering the existing literature, the ethical concerns discussed below have been regarded as key challenges in AI adoption in banking and financial contexts. The significance of this study highlights two major contexts: the generally known ethical issues and particular ethical issues in utilizing AI in the context of banking. The extended discussion below depicts the most widely concerned ethical issues in current literature in relation to ethical use of AI in financial or banking contexts.

3.2.1 Privacy challenges

Data privacy has become a significant concern in the current technological landscape due to the complexities and increasingly sophisticated means of accessing personal information, which is no longer solely individual but often indirectly shared (Kamleitner & Mitchell, 2019). As individuals are interconnected through various networks, their personal information becomes intertwined in a broader social context. Consequently, privacy has evolved into a multi-actor phenomenon, where individuals hold and passively share each other's information, sometimes unintentionally. Hence, it has become an issue when the sensitive data has been shared and used without consent.

In the context of AI adoption, data has emerged as a fundamental technical element. AI operates through three essential components: data gathering, data processing, and data output (Martin & Zimmermann, 2024). The most controversial aspect of AI-driven data handling is that it processes vast amounts of information, including both intentionally provided data and non-intentionally provided data, such as digital footprints left behind through routine activities, including digital recognition systems and online tracking mechanisms. Research has shown that certain uncontrollable factors undermine an individual's ability to rationally assess the decision to disclose personal data (Acquisti et al., 2020; Kamleitner & Mitchell, 2019; Martin & Zimmermann, 2024). From a broader perspective, it is sometimes inevitable that personal information is utilized by third parties without explicit consent. AI's capability to extract data from multiple sources further complicates this issue, raising concerns that the benefits of AI adoption may come at the expense of data privacy risks.

The banks and financial institutions, which handle vast amounts of highly sensitive customer data, are particularly vulnerable to privacy breaches. For instance, Large Language Models (LLMs) are increasingly used in chatbots and virtual assistants for customer interactions. As outlined in privacy policies, data is gathered through various channels, including direct user inputs, device and browser data such as IP addresses and locations, and even external sources such as social media and internet-synchronized databases (Mogaji & Nguyen, 2022). Given this extensive data collection, privacy risks in AI-powered banking operations are significantly higher compared to traditional banking systems.

To address these concerns, the European Union's General Data Protection Regulation (GDPR), which came into effect in 2018, has established mandatory requirements for businesses to ensure data protection and strengthen privacy safeguards for individuals (Kamaruddin et al., 2023). GDPR enforces stringent guidelines on personal data processing, thereby providing a regulatory framework that enhances data privacy protections.

Financial institutions that have hesitated to integrate AI technologies into their operations, often cite data privacy risks and the potential for discriminatory behaviour as primary barriers (Lazo & Ebardo, 2023). However, privacy concerns can be mitigated to some extent through effective regulatory frameworks, proper data processing protocols, and increased user awareness regarding per-

sonal data management. The principles of privacy by design and privacy by default play a crucial role in ensuring data confidentiality and minimizing potential harm (Milossi et al., 2021). Given the complexity of AI algorithms and logical frameworks, careful and strategic navigation is essential in designing AI-driven financial products that uphold ethical privacy standards.

3.2.2 Transparency challenges

Transparency is widely recognized as a fundamental factor in establishing trust within society. However, AI-driven decision-making is often characterized as a "black box" due to its inherent lack of transparency (Gaurav & Tiwari, 2023). Hanif et al., (2021) explain this to be an inherited characteristic of ML models which are highly beneficial in analysing and processing large amount of data within least amount of time. In the context of AI, transparency refers to the ability to understand how decisions are made, the rationale behind specific outcomes, and the nature of the data utilized in the decision-making process. Humans inherently possess curiosity and a desire to comprehend their surroundings. When transparency is lacking, this fundamental need for understanding is left unfulfilled, ultimately leading to a decline in trust.

Given the complexity and technical nature of AI, particularly for non-technical individuals, there is an increasing demand for explanations regarding the solutions and decisions generated by AI systems (Kurshan et al., 2022). In the banking sector, transparency is particularly significant, as customers seek to understand the reasons behind critical financial decisions, such as loan rejections or declined transactions. Customers perceive access to such information as a justified right, expecting banks to provide clear explanations for AI-driven outcomes. According to the study by Kumar, Sharma and Mahdavi (2021), since AI is extensively used for credit scoring, fraud detection, and risk management, the degree of transparency in its decision-making processes is a key concern for both customers and financial institutions. The ethical challenge arises when AI is unable to offer a specific and comprehensible justification for a decision beyond stating that it is based on patterns identified in training data.

AI models are increasingly utilized in credit scoring as part of the loan approval process (Botunac et al., 2024). However, even the developers of these models often struggle to fully explain how specific outcomes are generated, which factors exert the greatest influence on decisions, or whether these decisions are made in a fair and justifiable manner. While Explainable AI (XAI) and Interpretable Machine Learning (IML) aim to enhance the transparency of AI-driven decision-making, their practical implementation in the banking sector remains challenging due to technical constraints (Deloitte Insights, 2022). Consequently, the ethical concern surrounding the lack of sufficient transparency in AI-based decision-making is an overarching critical issue in the adoption of AI within the banking industry.

3.2.3 Algorithmic bias related challenges

Bias in AI, often referred to as algorithmic bias, is a widely recognized issue in AI models. Since these models rely on the data used for training, human biases or inaccuracies in the data can perpetuate and even amplify discrimination, stereotyping, and false-positive decision-making (Waller et al., 2025). Cheng, Varshney and Liu in 2021 describe that biases in training data may originate from existing social structures, such as racism, prejudice, and classism. Additionally, factors such as gender bias, socioeconomic bias, and unintended preferences in input data can contribute to ethical dilemmas and discriminatory outcomes. Bias in AI development can also persist through a continuous cycle, where a lack of diversity among development teams influences how models are trained, further embedding systemic discrimination throughout data processing and output phases. In post-training stages, amplified biases may evolve into new, even more discriminatory behaviours (Waller et al., 2025).

In the context of banking operations, Ridzuan et al. (2024) argue that bias must be eliminated in evaluating creditworthiness and determining access to financial services. Bias in AI-driven banking systems undermines fairness, potentially marginalizing certain groups and eroding trust in financial institutions. If bias is not addressed, it may jeopardize an institution's reputation and client confidence. Ideally, AI systems should surpass human decision-making in terms of fairness and impartiality. To achieve this, bias mitigation strategies must be implemented at both the training data and model development stages, ensuring that AI operates on accurate and unbiased data throughout the decision-making process.

Bias in AI adoption within the banking sector presents a significant ethical challenge, as the primary objective of integrating AI is to minimize human errors and enhance fairness in financial crediting and lending processes. However, research has demonstrated that AI-driven financial systems can inadvertently reinforce existing biases. For instance, a study by the National Bureau of Economic Research in 2019 found that AI-powered mortgage approval systems were more likely to deny minority applicants contrasted to equally qualified white applicants. The study also bared that Latinx and African American borrowers were subjected to higher interest rates, 5.3 basis points higher for purchase mortgages and 3.1 basis points higher for refinance mortgages (Chopra, 2024). Similarly, instances of unfair insurance pricing and biased investment recommendations from AI powered robo-advisors have emerged as ethical concerns in AI-driven banking operations (Barile et al., 2024). Addressing these challenges is crucial to ensuring that AI adoption in the financial sector aligns with principles of fairness and non-discrimination.

3.2.4 Accountability related challenges

Accountability in AI adoption encompasses a broad definition of the obligation to inform, justify, and take responsibility for one's conduct to relevant authori-

ties or regulatory bodies (Chopra, 2024). It is fundamentally relationship and delegation-based, grounded in established principles. Novelli, Taddeo and Floridi (2024) examine AI accountability from a broader perspective, highlighting that it has been imprecisely defined due to its multifaceted nature and the sociotechnical complexity of diverse AI system. The concept of accountability is critical in AI governance, ensuring that AI adoption is fair, and its consequences align with societal values.

Within the banking and financial sector, a sociotechnical approach is essential for AI development, from model design to deployment and utilization. Given the critical nature of decision-making in banking, accountability is paramount, as customers seek clarity on who holds authority over specific justifications. Furthermore, accountability fosters compliance, ensuring that AI utilization aligns with ethical, legal, and technical best. Most of existing studies suggest that establishing robust reporting mechanisms is crucial for explaining and justifying AI-driven decisions, while continuous monitoring is necessary for evaluating AI performance practices (Nabbosa & Kaar, 2020; Novelli et al., 2024). Legislative enforcement further strengthens accountability by defining the consequences of AI-related actions, ultimately contributing to responsible AI governance, risk mitigation, and public trust in AI-embedded financial systems.

Accountability becomes an ethical issue when no responsible party can be identified in cases where AI-driven decisions result in harm or injustice to banking customers or organization (Ridzuan et al., 2024). As financial institutions are traditionally governed by frameworks that ensure fair and justified interactions with clients, the introduction of AI necessitates a clear delineation of responsibility. This is particularly relevant in cases where financial losses occur due to AI-driven processes, requiring well-defined accountability mechanisms to ensure that losses are fairly distributed among responsible parties. Additionally, banks must navigate regulatory compliance when addressing customer complaints related to AI-based decisions. The ethical dimension of accountability and liability, therefore, presents significant complexities when applied to real-world financial operations.

Based on the analysis presented in the section on the ethical landscape of AI adoption, it is evident that while a substantial body of literature exists on AI adoption in banking. Through this review, I have identified key thematic categories of ethical concerns, which will form the basis for the final synthesis of findings in this SLR. Notably, the existing literature rarely problematizes the conceptualization of ethics or ethical behaviour in depth. However, this preliminary overview serves as an important role by laying the groundwork for the extrapolation of interrelated dimensions from the literature. These dimensions are not presented as isolated in the current body of research but rather as overlapping and interconnected ethical categories. This framework will later be used to analyse how these ethical challenges influence the broader discourse and practical implementation of AI in banking.

4 METHODOLOGY

The research methodology employed in this study is the Systematic Literature Review (SLR), a scientific and structured approach to reviewing existing literature. SLR is a rigorous and comprehensive method designed to identify, evaluate, and synthesize all relevant research on a specific topic or research question (Boell & Cecez-Kecmanovic, 2014, 2015). Unlike a single case or comparative case study which provides deep but context-bound insights, an SLR synthesizes findings across multiple contexts (such as different banks, geographic regions and technical approaches) to reveal broader patterns in how ethical challenges of AI have been conceptualized and addressed. Surveys or questionnaire - based studies, by contrast, can capture perceptions at a single point in time but cannot systematically trace the evolution of academic discourse, nor can they correct for publication bias or methodological heterogeneity; the SLR's explicit inclusion/exclusion criteria, multi-database search strategy, and screening steps ensure that the literature sampling is both transparent and unbiased. Experimental or design-science approaches may demonstrate how a particular mitigation tool or ethical framework performs in practice, but they lack the capability to map the full landscape of existing solutions, identify latent theoretical gaps, or benchmark emerging practices against one another. By aggregating 72 peer-reviewed studies spanning 2018-2025, this SLR not only provides a holistic and theory-driven categorization of ethical challenges but also establishes a reproducible roadmap for future researchers to update or extend this work as AI ethics continues to evolve.

To ensure rigor and transparency, a comprehensive search strategy was implemented, identifying peer-reviewed journal articles and conference papers from established academic databases. A clear search criterion, detailed later in the text, guided the selection process. The review adheres to strict inclusion and exclusion criteria, ensuring a focused and relevant study. Furthermore, as systematic literature reviews must be reproducible, the entire process has been carefully documented for future research and validation using PRISMA diagram (*PRISMA Statement*, 2020). The temporal scope of this study spans 2018–2025, reflecting the significant increase in academic literature on AI ethics in

banking during this period. The selected timeframe ensures that the study captures recent advancements, emerging challenges, and evolving discussions in the domain.

4.1 The search strategy of SLR process

The selection of literature plays a major role in SLR process, the formulation of search string was based on the key words in research area. Despite defining a structured search string query, a major challenge was to deal with the irrelevant studies which actually related to AI but not explicitly belong to the banking sector or not discussing any ethical issues. This challenge has been overcome with manual screening while using the automated tool, Ryyan (Ryyan, 2025). Another challenge was that sometimes databases resulted with flooding of literature in the search domain and had to use temporal scope from the inclusion/exclusion protocol to narrow down the search into specific number of outputs. So, the temporal filter had to be applied to the initial search phase to obtain a certain number in the results to be documented in PRISMA.

4.1.1 Digital resources and databases

The selection of databases to be used in the research was based on the quality of the articles published as per the categorized standards by the academic experts (Petersen et al., 2015) which elaborates the importance of selecting quality secondary data from the databases depending on quality criteria. The list of databases used for academic article retrieval is given below.

- ACM Digital Library
- Emerald Insight
- IEEE Xplore
- ScienceDirect
- Scopus

Other information retrieval using online digital resources were mainly used for depicting the current landscape of AI adoption reports or surveys. All the sources were cited in the text to ethically use them in the thesis.

4.1.2 Search phrases and keywords

An extensive keyword search was carried out to collect the most relevant articles. The keywords and search criteria used in collecting quality articles from databases which are widely used in the study domain. The focus was to select journal articles, hence other conference papers, books, book chapters, opinion papers were excluded from the search. Within the last few years, large numbers

of articles have been published in various sub disciplines related to AI. Keyword search was targeting on the title, abstract and keywords to identify the most relevant articles for the study domain. The initial search hit thousands of papers hence the time frame had to be applied as a filter to sort out the most relevant choices. The filters are described in the inclusion and exclusion criteria in more detail.

Moreover, the determination of keywords was challenging due to the extensive availability of research articles in the domain of AI in varied disciplines. Depending on the requirement of databases, search was carried out in title and the abstract and in full text and metadata.

TABLE 1 Keywords and search strings

Data base	**Search Strings**
ACM Digital Library	Fulltext:(AI) AND Fulltext:(ethic*) AND Abstract:(bank*) AND AllField:("Artificial Intelligence") AND Abstract:(financ*)
Emerald Insight	title:"Artificial Intelligence" OR (title:"AI") AND (title:"bank" OR "banking") AND (title:"ethic" OR "ethical") AND (abstract:"financial" OR "finance")
IEEE Xplore	("Full Text & Metadata":"AI" OR "Full Text & Metadata":"Artificial Intelligence") AND ("Full Text & Metadata":ethic*) AND ("Full Text & Metadata":bank*) AND ("Full Text & Metadata":financ*)
ScienceDirect	TITLE-ABS-KEY("artificial intelligence" OR "AI") AND TITLE-ABS-KEY("banking" OR "bank" OR "financial") AND ABS-KEY("ethical challenges" OR ethics))
Scopus	(TITLE-ABS-KEY (artificial AND intelligence) AND TITLE-ABS-KEY (ethic*) AND TITLE-ABS-KEY (bank*) OR TITLE-ABS-KEY (fiananc*))

As shown in Table 1, the primary search keywords included "artificial intelligence," "AI," "ethic," "bank," and "financial." Additionally, search strings were modified based on the requirements of each database. For instance, the asterisk (*) wildcard was used to include variations of a word, e.g., "ethic*" retrieved results for ethics, ethical, and ethically.

Some databases, such as Emerald Insight, initially returned over 1,000 records without any filters applied. In such cases, time duration (2018–2025) and access type (open access) were used as filters to refine the results and document them appropriately in the PRISMA framework.

4.2 Inclusion and exclusion criteria for eligibility

The initial selection of the study materials required inclusion and exclusion criteria based on the requirements of the study. The eligibility criteria consisted of timeline, type of document, language, access permission and relevance to the study.

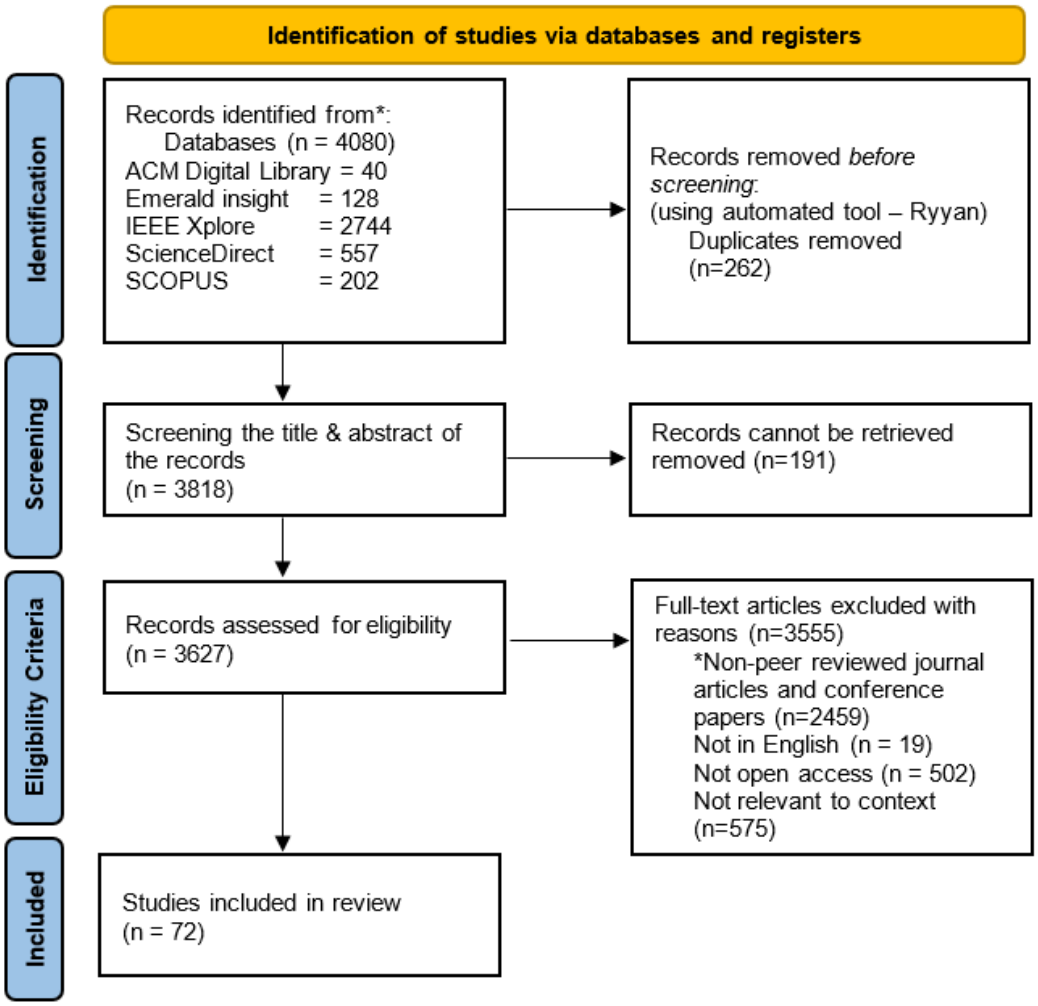
TABLE 2 Inclusion and exclusion criteria

Selection criteria	**Inclusion**	**Exclusion**
Timeline of publication	January 2018 – January 2025	Before January 2018 and after January 2025
Document type	Peer-reviewed Journal articles and conference papers	Non-peer reviewed articles and conference papers, books, chapters, magazines and … etc.
Language	English	Non-English
Access type	Open Access	Other access permissions
Relevance to context	AI and ethical challenges in banking sector related	Not relevant to AI and ethical challenges in the banking sector

4.3 Literature screening and selection process

Given the vast number of publications on AI, a rigorous screening process is essential to identify the most relevant and high-quality studies that effectively address the research questions. A key challenge in this process is the removal of duplicate records, as many articles appear across multiple databases. Eliminating these duplicates significantly reduces the initial dataset, refining the pool of selected studies. To streamline and automate this process, the Rayyan tool (Ryyan, 2025) was utilized to identify and remove duplicates efficiently. Additionally, Rayyan facilitated manual abstract and keyword screening to assess whether the selected studies aligned with the research scope. However, it was observed that some articles, despite containing relevant keywords in their title or abstract, lacked substantive discussion on the specified research area.

To ensure a high level of relevance and quality, the final step comprised a full-text review of the shortlisted articles. This comprehensive evaluation ensured that the selected studies provided meaningful insights to answer the research questions (RQs) and contribute effectively to the synthesis of findings. The systematic review process is visually represented in the PRISMA diagram, which illustrates the flow of study selection from initial identification to final inclusion.



*All the other types of articles such as non-peer reviewed articles and conference papers, magazine articles, books, chapters, proceedings, newsletters, ... etc.

FIGURE 2 PRISMA 2020 flow diagram for the study

As Figure 2 illustrates the initial number of records (4,080) retrieved after applying filters for open access and a time frame of 2018–2025 during the initial keyword search. These two filters were applied to manage the overwhelming number of search results, ensuring a more focused and relevant dataset. Following this, 262 duplicate articles were identified and removed using the automated tool, Rayyan.

During the screening process, title and abstract screening was conducted for the remaining 3,818 records using Rayyan. However, 191 records had to be removed due to access restrictions, preventing retrieval.

Next, the inclusion and exclusion criteria were applied to 3,627 records, leading to the exclusion of 3,555 articles for the following reasons: Since this study focuses solely on peer-reviewed journal articles and conference papers, all other publication types such as non-peer-reviewed articles or conference papers, magazine articles, books, book chapters, proceedings, and newsletters were excluded (2,459 records). Non-English articles and articles behind pay-

walls (without open-access availability) were also removed. During the title and abstract screening, 575 articles were excluded due to irrelevance to the study context.

The final number of studies included in this study was 72. These 72 articles consisted of 37 peer-reviewed conference papers and 35 journal articles. The total number of articles were considered for categorization of ethical challenges and in the final synthesis of literature according to category. This study includes both peer-reviewed conference papers and journal articles in the final synthesis to ensure a comprehensive and balanced perspective. While journal articles typically offer extensive theoretical grounding, detailed methodologies, and robust empirical analyses, conference papers often present emerging ideas, novel frameworks, and preliminary findings that reflect the latest developments in the field. Despite differences in depth, rigor, and longevity, conference papers contribute valuable insights, particularly in fast-evolving domains such as AI ethics, where timely discussions and cutting-edge explorations are crucial. Therefore, integrating both sources enriches the synthesis by capturing both established knowledge and emerging trends, ultimately providing a more holistic understanding of the subject matter. The study aims to build a robust analysis of ethical challenges in AI adoption in banking, requiring sources with higher validation and reliability. The selected articles were chosen based on their credibility, reliability, and well-defined research objectives. Each study employs a clear methodological approach and presents results supported by substantial evidence. Ensuring rigor and relevance, all 72 articles meet the pre-established inclusion criteria and represent high-quality academic contributions in the domain.

A full-text review was then conducted to assess the relevance, quality, and applicability of these articles for analysis. In a SLR, the quality of the selected studies holds greater significance than the quantity of literature reviewed. For efficient organization and citation management, the selected studies were stored in the Zotero reference management system (Zotero, 2025). Additionally, all bibliographic records were systematically categorized and presented in Appendix 1, based on the predefined inclusion and exclusion criteria to maintain structured documentation throughout the research process. OpenAI's ChatGPT-4 was utilised for refining the text, with the aim of ensuring clarity, coherence, and error-free academic documentation.

The process of extracting significantly important and meaningful data from diverse sources has proved to be time consuming. On the other hand, ethical challenges in AI adoption in the financial or banking sector have been often implicitly discussed rather than explicitly discussed. Hence, interpretation of results must be considered based on underlying theoretical lenses and perspectives. This process required critical thinking to compare, contrast and synthesize the findings while trying to focus on the objectives of own study and avoiding possible biasness.

5 THE ANALYSIS OF FINDINGS

This section presents the final synthesis of results derived from the SLR on the ethical challenges associated with the adoption of AI in banking operations. To facilitate this analysis, a conceptual model has been developed which offers a systematic framework, capturing the interdependencies and interrelationships among key dimensions. Further, this categorization of ethical challenges leads to a framework which provides a comprehensive lens through which the ethical implications of AI adoption in the banking sector can be more effectively understood and addressed.

Considering the theoretical influence on the analysis, from the perspective of UTAUT, key constructs were applied to assess customer satisfaction, trust, and reputation in relation to AI adoption. These factors were instrumental in evaluating technology acceptance, customer concerns regarding AI-driven decision-making, and the overall societal and technological impact of AI implementation in banking. This theoretical lens provided insights into how AI adoption influences customer perceptions, the extent to which AI-driven systems facilitate or hinder customer interactions, and the role of social influence in shaping trust in AI-based financial services.

Utilitarianism was employed as a theoretical lens to examine whether the benefits of AI adoption in banking outweigh its ethical risks. This included an assessment of whether AI-driven financial innovations maximize overall utility by enhancing efficiency while minimizing ethical harms. The analysis further explored whether AI prevents or exacerbates ethical risks such as bias, discrimination, and privacy violations. By applying this ethical perspective, the study critically evaluates the extent to which AI adoption aligns with broader societal welfare and organizational ethics.

Virtue Ethics provided an additional ethical dimension by assessing how financial institutions uphold ethical responsibilities in AI-driven decision-making. This theoretical perspective facilitated an examination of how AI models embody ethical values, the extent to which organizations demonstrate accountability when AI-based decisions lead to adverse consequences, and the role of ethical corporate behaviour in fostering responsible AI adoption. This

perspective helped in identifying the ethical expectations placed on banking institutions and their role in ensuring fairness and integrity in AI deployment.

Actor-Network Theory was employed to analyse the socio-technical interactions between AI-driven decision-making systems, human oversight, and regulatory frameworks. This lens was used to investigate whether AI models function as independent decision-makers within a socio-technical network or whether human intervention remains a crucial element in ethical governance. Additionally, ANT provided insights into the interplay between financial institutions, regulatory bodies, and AI systems, highlighting the necessity of human oversight, regulatory compliance, and ethical accountability in AI-driven banking operations. In Reassembling the Social, Latour (2005) distinguishes between actors, entities that actively carry interests and have the capacity to transform the state of affairs and intermediaries, which merely transport meaning without modifying it. Applying this distinction to the context of this study, various human and non-human elements within the banking ecosystem function as either actors or intermediaries in the ethical adoption of AI. For instance, the peer-reviewed research articles analysed in this systematic literature review may be considered intermediaries in that they convey messages, arguments, and findings related to the ethical use of AI. However, they can also act as actors when they influence regulatory bodies, policy makers, AI developers, or academic readers, thus shaping attitudes, policies, and practices surrounding responsible AI adoption. This dual role underscores Latour's (2005) assertion that actors are defined not by their inherent properties but by the actions they enable within the network. Moreover, ANT's concept of translation, the process by which actors interpret and reframe meanings to align with their own goals is particularly relevant to this study and will be discussed further in subsequent sections. It provides a useful lens to examine how diverse actors within the socio-technical banking network such as banks, customers, legal frameworks, and technological artefacts negotiate and reshape the understanding and implementation of ethical AI.

The theoretical models guide the interpretation of findings, ensuring a structured and comprehensive approach to understanding the ethical implications of AI-driven financial systems. The final synthesis is organised into three main sections. The first section, addressing RQ1, outlines the ethical challenges identified in the existing literature concerning AI implementation in banking. The second section responds to RQ2 by introducing my categorisation of these ethical challenges, highlighting the interconnections among the identified categories. The third section addresses RQ3, through an analysis of the methodological approaches employed in the reviewed studies, offering insights to inform future research, particularly in problematising complex issues such as the ethical development and deployment of AI.

5.1 Analysis of ethical challenges of AI adoption

The synthesis is based on a review of 72 relevant studies, selected to answer the three research questions. Findings related to RQ1 reveal the most frequently discussed ethical challenges in the context of AI adoption in banking and financial services. These results are summarised in the table 3 below (detailed table in Appendix 1). Data privacy and security emerged as the most prominent concern, discussed in 61 out of the 72 studies, reflecting the pervasive issues surrounding the handling of sensitive financial information. Bias and discrimination were highlighted in 51 studies, while transparency in AI decision-making processes was addressed in 46 studies. Furthermore, the accountability and responsibility for AI-driven decisions were discussed in 40 studies. Beyond the primary concerns, additional ethical challenges were identified. Fairness in ensuring equitable treatment across all customer groups, was discussed in 18 studies. 16 studies raised concerns about the lack of human involvement and the threat of job displacement caused by automation. Moreover, the importance of explainability in AI decision-making processes appeared in 5 studies. The issue of lacking empathy in AI systems was mentioned in 2 studies.

TABLE 3 The key ethical challenges of AI adoption

Author, year	Data privacy & security	Bias & Discrimination	Transparency	Accountability	Fairness	Human displacement	Explainability	Empathy
Total #	61	51	46	40	18	16	05	02

These findings underscore the answer to RQ1: four major ethical concerns dominate the literature on AI adoption in banking. Firstly, data privacy and security, primarily driven by the vast amounts of data required for AI technologies, stands out as the most significant concern. Secondly, the opaque or "black box" nature of AI algorithms has raised concerns about potential bias and discrimination, the transparency of AI decision-making, and the accountability for outcomes, collectively forming the next three major ethical challenges.

In addition, several secondary concerns including unfair decision-making processes, the displacement of human labour by AI systems, insufficient human intervention, the limited explainability of AI decisions, and the absence of empathy and emotional engagement with customers were noted. While these challenges appear less frequently in the reviewed literature, they nonetheless represent important dimensions of the ethical discourse surrounding AI in banking.

5.2 Categorization of ethical challenges of AI adoption

In this section, my conceptual model for the categorisation of ethical challenges is presented. Drawing from the existing literature, the major ethical challenges associated with the adoption of AI in banking have been identified. The synthesis of findings leads to a conceptual framework, that not only categorises these ethical challenges but also illustrates their interrelated nature and the ways in which they influence one another.

It became evident during the analysis that certain ethical challenges are having strong intensity upon others whereas others show a lower intensity. This is largely due to the complexity of problematising these concerns in a universally applicable way. Each category within the framework is further divided into sub-categories, reflecting the multidimensional character of these ethical issues and their interdependencies across various perspectives.

The proposed framework is informed by ethical theories and is designed to explore the challenges associated with the adoption of AI in banking operations, with particular attention to their multifaceted impact on the diverse actors within the ecosystem. This analysis can be further enriched through the lens of ANT, which conceptualizes both human and non-human entities as actors engaged in dynamic and interconnected networks. According Callon M. (1986), these actors form and maintain networks through processes of translation, wherein information, interests, and values are interpreted and rearticulated to align with the goals of various stakeholders. Each network develops its own unique translation mechanisms, which can generate tensions due to differing interpretations, interests, and power dynamics. These tensions shape the influence and agency of actors within and across networks, ultimately affecting how AI technologies are adopted, accepted, or resisted. This theoretical framing thus provides a nuanced understanding of the complex, evolving relationships among actors in the banking ecosystem, and highlights the ethical implications of AI integration as situated within these interdependent networks.

Furthermore, the framework acknowledges that the implications of AI adoption are not uniform across all stakeholders. Consistent with the UTAUT, the analysis recognizes that different users, whether individuals, organizations, or wider societal actors, experience the impacts of AI in varied ways. Some challenges predominantly affect organizations, while others have more profound implications for customers or external stakeholders, and vice versa.

These categories or dimensions are not explicitly delineated in the existing literature; rather, being synthesized and extrapolated them through an in-depth review of relevant studies. This process enabled the construction of a novel framework that visualizes the interrelated and coherent dimensions of ethical challenges associated with AI adoption within the banking sector.

1. Ethical challenges of AI adoption in banking in general perspective (implicit)
2. Ethical challenges of AI adoption in banking in technical perspective
3. Ethical challenges of AI adoption in banking in legal and regulatory framework perspective
4. Ethical challenges of AI adoption in banking in explicit perspective

This conceptual framework acknowledges that these domains are not isolated but rather dynamically interact. To enhance the depth and applicability of this analysis, this study aims to map the key relationships between these categories, highlighting the "ethical relations" inherent in AI adoption. By identifying these interconnections, the study seeks to demonstrate how technical and functional AI integration must be refined to ensure the development and deployment of ethically sound and responsible AI systems in the banking sector.

Detailed below is a summary of my own synthesis of the ethical challenges categorized according to the study's research objectives.

TABLE 4 Author's own categorization of ethical challenges according to study objectives

Categorization of ethical challenges of AI adoption in banking	**Sub categorization**
Ethical challenges of AI adoption in banking in general perspective (implicit)	A: Ethical challenges of AI adoption in banks (Implicit) B: AI adoption in specific areas in banking operations; customer service, mobile banking (General) C: AI for fraud detection D: AI for credit scoring E: AI for money laundering detection F: AI for debt recovery
Ethical challenges of AI adoption in banking in technical perspective	A: AI adoption in banks for cybersecurity enforcement B: Explainable AI (XAI) adoption in banking operations C: Machine Learning (ML) adoption in banking operations D: Biometric analysis with AI adoptability in banking
Ethical challenges of AI adoption in banking in legal and regulatory framework perspective	A: AI adoption in banking with relevance to GDPR B: AI adoption in banking with relevance to other legal frameworks
Ethical challenges of AI adoption in banking in explicit perspective	A: Data privacy and security challenges B: Accountability and liability C: AI system development challenges D: Other ethical challenges (fairness, explainability, trust, empathy, human displacement concerns and …etc.)

5.2.1 AI adoption in banking (general perspective)

The adoption of AI in banking presents numerous advantages, particularly when analysed through the lens of the UTAUT. Core constructs such as performance expectancy, effort expectancy, social influence, and facilitating conditions explain how various stakeholders such as customers, employees, and institutions interact with AI technologies. Customers benefit from improved transaction efficiency and service accessibility, while organizations gain enhanced predictive capabilities and operational agility. Effort expectancy underscores the ease of using AI-driven tools like mobile apps and robo-advisors. Social influence highlights the role of stakeholder interaction in promoting trust and competitive positioning, whereas facilitating conditions include technological infrastructure and customer support systems that drive AI adoption.

When viewed in conjunction with ANT, each actor, be it customer, financial institution, regulator, or AI technology interacts within this network, shaping and being shaped by others. These interdependencies lead to varying interpretations and ethical evaluations, particularly concerning data privacy, bias, transparency, and accountability.

Despite these theoretical insights, the SLR conducted for this study found that many existing studies discuss ethical challenges only superficially. Nearly half of the reviewed literature addresses concerns like data privacy and fairness without engaging in deep analysis or problematization. This superficial engagement limits the field's ability to develop meaningful insights or standardized ethical guidelines. One reason for this gap is the inherent complexity and rapid evolution of AI technologies, which complicates ethical evaluations and regulatory responses. As a result, this study establishes a distinct category for literature that acknowledges ethical issues at a general level, highlighting the need for more critical and detailed future research into ethical AI adoption in banking.

In this analysis, I have identified and sub-categorized the discussed ethical issues related to AI adoption by focusing on the specific areas covered in the literature. (As per the tables given in Appendix 2) To this end, six main subcategories have been established to classify the existing studies:

A: Ethical challenges of AI adoption in banks (Implicit)
B: AI adoption in specific areas in banking operations; customer service, mobile banking
C: AI for fraud detection
D: AI for credit scoring
E: AI for money laundering detection
F: AI for debt recovery

These six sub-categorizations facilitate a systematic analysis of how ethical challenges are discussed across different applications of AI in the banking sector, thereby contributing to a nuanced understanding of the domain.

A: Ethical challenges of AI adoption in banks (Implicit)

A total of 14 peer-reviewed journal articles and 7 peer-reviewed conference papers; altogether 21 studies have been classified within this sub-category. The authors of these studies acknowledge that ethical concerns in AI adoption within banking operations must be carefully addressed; however, their treatment of these issues remains relatively superficial. Below are selected quotes from various studies in this sub-category, which serve as evidence supporting the logical approach employed in this analysis.

Mohsen, Hamdan, and Shoaib (2024) identify ethical risks associated with predictive AI models, particularly concerning data security, bias, and potential inaccuracies in decision-making. However, the study mainly outlines these issues without deeper examination of their systemic implications. Similarly, Mogaji and Nguyen (2022) studies about a subset of actors in the ecosystem according to ANT; the managers who must bear the responsibilities of banking decision-making up to some extent, through interviews with bank managers, reveal managerial apprehensions about ethical considerations when working with developers and suppliers, but stop short of thoroughly exploring how these relationships shape ethical outcomes. Moreover, Eskandarany (2024), focusing on Saudi Arabian banks, notes that board members emphasize transparency and fairness in AI systems, quoting "*We cannot afford to have biased algorithms or opaque decision-making processes. As board members, we are responsible for ensuring our AI systems are transparent and fair, especially given the sensitive nature of financial data*". Durongkadej et al. (2024) echo these concerns, citing real-world incidents like the Wells Fargo mortgage error to illustrate the potential harm of AI-driven decisions. Nonetheless, the study largely catalogues risks such as lawsuits, bias, and privacy threats without delving into the underlying causes or mitigation strategies. Hence, different stakeholders of this technology adoption have been impacted by different ethical challenges in their own perceptions.

Fundira et al. (2024) offer insights from end users; the major actors representing the network, who express concerns over the loss of human touch and trust in AI systems by quoting *"AI has left a void of human touch…"* which in their viewpoint is an ethical concern related to trust. Moreover, a response said "*… industry concerns about fairness, transparency, reliability, and privacy affecting customer trust*". Issues are mentioned, but the analysis remains at the level of user perception rather than systematically unpacking these ethical dimensions. Similarly, Sadok et al. (2023) and Mehndiratta et al. (2023) both discuss the ethical risks of AI's reliance on vast amounts of personal data, particularly from social media and financial records. Sadok explains saying, "*… there are also others of an ethical nature following the use of big data from social networks. The heart of the problem is to know if the customers would consent to some of their personal data being used during their relationship with the bank or the insurance company*". However, their discussions tend to generalize the challenges, emphasizing the need for customer consent and data protection without providing an in-depth ethical evaluation.

Collectively, these studies recognize the importance of addressing ethical concerns in AI adoption and advocate for the establishment of legal frameworks. However, their treatment of ethical issues is often superficial, focusing primarily on identifying risks rather than offering critical, detailed analyses of how these risks manifest, interact, and can be meaningfully mitigated within the complex banking ecosystem. As such, there remains a significant gap in the literature regarding comprehensive, in-depth exploration of ethical implications in AI-enabled banking practices.

B: AI adoption in specific areas in banking operations; customer service, mobile banking (General)

5 peer-reviewed journal articles and 5 peer-reviewed conference papers; altogether 10 studies consider AI adoption in banks specializing in specific areas of banking operations. In instance; personalized banking, mobile banking and service platforms, robo-advisory for customers, chatbots and virtual assistants for consumer banking, customer relationship management with AI models… etc. as a whole the actors of the banking ecosystem according to ANT, can be named as areas which have been explored by the selected articles.

Within the domain of personalized banking and customer service, multiple studies by various scholars highlight that AI adoption introduces significant ethical concerns, particularly regarding data privacy, security, transparency, and accountability. One major concern raised by the study by Gigante & Zago, (2023) is the fundamental ethical dilemma surrounding AI's replication of human intelligence; *"AI involves ethical issues, too: to what extent is it appropriate to recreate human intelligence, assigning it to machines?"*. This concern is further compounded by customer trust issues, as banking services involve handling highly sensitive personal data: "*concerns emerge about security and privacy. Because this service is offered through the collection of sensible data, customers should trust the provider and feel protected from data breaches*". Another study on value co-creation with AI in mobile banking platforms by Manser Payne et al., (2021) further highlights ethical concerns related to the lack of human interaction in AI-driven services; *"AI service delivery may offer utilitarian (i.e. functional) value by performing secure banking activities… However, for customers with a higher need for human interaction, service delivery is dependent on employee–customer interaction"*. This highlight facilitating conditions in UTAUT, that certain users need human intervention to accept the technology adoption. While the study acknowledges that AI integration in mobile banking can enhance service value, it also emphasizes that ethical concerns related to data usage remain unresolved.

Additionally, an empirical study on personalized banking experiences by Rohella et al., (2024) underscores the role of customer diversity in shaping the ethical challenges of AI adoption. Considering the ANT perspective, the study highlights that certain consumer groups, particularly vulnerable populations, may face heightened risks associated with AI-driven financial services; "*Many consumers, mainly those belonging to vulnerable groups, have legitimate reasons for worrying about their data being used without permission or in ways that they do not understand.*". The study also stresses the potential for AI to compromise cus-

tomer privacy and security, which could lead to regulatory and compliance challenges; "*AI can adversely impact customer privacy and security, which can lead to compliance and legal issues*".

There is a growing trend of employing robo-advisory platforms powered by AI technologies to enhance customer experience and provide personalized banking services. A qualitative study by Barile et al., (2024) analyzing seven robo-advisory (RA) platforms in the banking sector highlighted several ethical concerns, stating, *"The absence of human contact can be a weakness in RA platforms,"* and *"Relying solely on algorithms and not including any services involving human interaction in a company model is inadequate to meet the requirements of customers in decision-making"*. Another study based on data from Poland by Piotrowski and Orzeszko, (2023) reveals that the critical importance of ethical compliance and human intervention within robo-advisory systems, noting, *"Consumers' willingness to use robo-advisory in bank services is strongly dependent on the honesty of financial advice and the transparency of the product offered"*. They further state that *"Trust in banks was not found to be statistically significant... however, the ethical aspects of how banks use personal data were highly important"*. This reveals that there is a significant requirement to safeguard customer data in all the AI adoptions in the banking operations.

These studies collectively emphasize the importance of addressing ethical considerations in AI-driven banking services, particularly concerning trust, data privacy, human interaction, and equitable access. However, despite acknowledging these concerns, there remains a gap in providing concrete solutions for mitigating these ethical risks.

C: AI for fraud detection

Two peer-reviewed conference papers were identified in this sub-category, focusing on fraud detection which is a critical banking operation encompassing areas such as credit card fraud, transaction fraud, and fraud targeting customers by third parties. One study by Johora et al., (2024) highlights that "*ethical considerations are foundational in AI-driven fraud prevention, particularly concerning transparency, accountability, and regulatory compliance*". The study emphasizes that AI adoption facilitates the effective detection and prevention of fraudulent activities while underscoring the importance of addressing ethical challenges. Another bibliometric analysis by Tiwari et al., (2023) examines the ethical concerns associated with AI-driven fraud detection technologies. It states, "*Ethical considerations and data privacy must be central to the deployment of AI in fraud detection, emphasizing fairness, transparency, and compliance with regulations*". These studies identify key ethical issues according to utilitarianism theories including data privacy, transparency, explainability, bias mitigation, and informed consent, and concludes by recommending regular monitoring and auditing to mitigate potential vulnerabilities.

D: AI for credit scoring

The ethical challenges associated with AI adoption in banking for credit scoring in the lending process have been examined in 3 peer-reviewed journal articles and a peer-reviewed conference paper within this study's sub-category. One study by Königstorfer & Thalmann, (2020) emphasizes that algorithmic bias in credit scoring AI models can be malicious, stating, *"Using biased data in the AI training process might cause the AI to make the same mistakes as previous human decision-makers."*. Moreover, it warns about the negative impact of ethical issues on banking operations, noting, *"If employees perceive the AI as a violation of their privacy, they might intentionally avoid using tools that incorporate AI."* This has a direct impact according to UTAUT, that perceived technology adoption is declined at the presence of ethical challenges by majority of users in the banking ecosystem. Additionally, the complexity of AI-driven decision making, often characterized as a "black box," is identified as a significant drawback. A similar study by Kumar et al., (2021) that focuses on the application of ML for credit scoring in rural finance, reports that *"There may inevitably be the occurrence of digital discrimination if personal data like age, gender, race, income, culture, religion, location, etc., have not been properly trained."* This study further underscores the importance of unbiased data by noting, *"ML-based models which are derived from historical decisions by the loan officers may also result in bias while extending loans,"* and concludes illustrating the complexities involved in adopting AI for credit scoring.

Highlighting the need for responsible AI in credit scoring , a study by Mathen & Paul, (2025) claims that *"Bias is mostly caused by feature engineering and variable selection… while such discriminations may not be illegal, they are ethically incorrect."* An architecture for a real-time classification model for loan credit approval using deep learning algorithms by Abakarim et al., (2018) underscores the challenge of creating effective systems, remarking that "*The challenge is to know how to build a proactive, powerful, responsible and ethical exploitation of personal data, to make loan applicant proposals more relevant and personalized*".

The studies in this sub-category indicate that the successful and effective adoption of AI in credit scoring is contingent upon the use of training data that is free from bias across factors such as gender, age, nationality, and other pertinent variables. Moreover, it is critical that the data used does not reflect historical inaccuracies stemming from human bias. This underscores the necessity for rigorous data curation, preprocessing, and continuous bias auditing to ensure that AI models in the lending process operate fairly and ethically. Additionally, integrating fairness metrics and validation procedures into model development can further enhance the reliability and ethical integrity of AI-driven credit scoring systems.

E: AI for money laundering detection

A peer-reviewed journal article and a peer-reviewed conference paper have been discussed in this sub-category regarding ethical challenges in financial crime prevention. Financial crimes remain a critical concern in the banking sector, and as technology advances, digital criminals have evolved correspondingly. This evolution underscores the significant need for effective financial crime prevention measures. In this context, AI adoption is vital for banks. Traditionally, banks have employed Anti-Money Laundering (AML) frameworks to detect and prevent illegal financial activities; however, AI offers a more dynamic analytical framework for the AML process. Despite its potential, ethical issues continue to pose barriers.

Study by Balaji, 2024 highlights these challenges by stating, "*AI adoption in AML/CFT, noting concerns about regulatory compliance, data privacy, and the ethical implications of BI-Enhancing Banking Transactions Through AI&ML Tools for AI-driven decision-making in financial institutions*". Chitimira et al., (2024), examining how AI technologies can be effectively used to detect and combat money laundering in the South African banking sector, emphasizes the importance of data precision to avoid ethical challenges and incorrect decision-making in the AML process. It states, *"If the collected data is biased or incomplete, the AI systems are likely to generate inaccurate or incomplete results which would ultimately lead to false positives or negatives"*.

Collectively, these studies indicate that AI represents a promising advancement in the complex domain of AML prevention in the banking and finance sectors, which is underpinned by robust legal frameworks. Nevertheless, ethical issues must be adequately addressed to ensure the responsible adoption of AI in this context.

F: AI for debt recovery

A peer-reviewed journal article and a peer-reviewed conference paper have been discussed in this sub-category related to debt recovery. Debt recovery represents another critical banking operation that traditionally requires rigorous and time-consuming processes. The integration of AI technologies for debt recovery introduces ethical challenges that must be carefully addressed. Study by Dogra & Kumar, (2024) emphasizes, "*Automating debt recovery processes raises ethical questions surrounding fairness, transparency, and accountability. While robots offer efficiency gains, they must operate within ethical boundaries, ensuring that decisions are made equitably and without bias*". This study expresses the necessity of overcoming ethical challenges in the implementation of AI-driven debt recovery processes to ensure that efficiency improvements do not compromise ethical standards.

5.2.2 AI adoption in banking in technical perspective

The analysis of existing literature reveals that the number of studies published in major academic databases that provide an in-depth technical analysis or propose AI-related frameworks in the banking sector remains relatively low. One possible explanation for this trend is the rapid evolution of AI-related technologies, which necessitates a high level of technical expertise to conduct specialized research in these emerging areas. As AI technologies continue to advance, the complexity of technical implementations increases, posing challenges for researchers seeking to develop comprehensive frameworks that address both technical and ethical considerations.

A notable exception in the literature is the cybersecurity domain, where AI applications have gained significant traction. A substantial body of literature focuses on AI-driven cybersecurity solutions, often incorporating highly technical discussions on implementation and performance. However, according to the ANT perspective, the banking ecosystem, comprising both human and technological actors, is also vulnerable to malicious attacks orchestrated by non-human agents leveraging advanced AI, often by dark actors. This vulnerability necessitates that regulators and other stakeholders demand the robust implementation of ethically sound and secure AI technologies in the banking sector.

Despite the prominence of cybersecurity studies, most of these works primarily emphasize technological and security aspects rather than addressing the ethical implications of AI adoption. Consequently, such studies fall outside the scope of this research, as they do not directly engage with the ethical concerns associated with AI implementation in banking. The most technically detailed literature within the selected corpus predominantly focuses on areas such as cybersecurity applications, the use of explainable AI (XAI) in financial services, machine learning (ML) applications in banking, and AI-based biometric analysis. While these studies contribute valuable insights into the technical dimensions of AI adoption, they often lack a direct examination of ethical considerations, regulatory compliance, and accountability mechanisms. To bridge this gap, the gathered literature analysed through a comprehensive and in-depth evaluation, with a particular focus on identifying and examining the ethical implications and interrelations associated with AI's technical adoption in banking. This approach ensured that the research not only acknowledges technical advancements but also critically explores their ethical ramifications within the financial sector. The subcategories are detailed as follows:

A: AI adoption in banks for cybersecurity enforcement
B: Explainable AI (XAI) adoption in banking operations
C: Machine Learning (ML) adoption in banking operations
D: Biometric analysis with AI adoptability in banking

A: AI adoption in banks for cybersecurity enforcement

Cybersecurity has emerged as a critical requirement in the current digital era, particularly in light of the malicious misuse of technology, including AI for exe-

cuting cyber-attacks, threats, and crimes (Eskandarany, 2024). With the increasing volume of big data and accelerated digitalization, AI has become a pivotal tool for mitigating cyber threats in financial institutions. For instance, machine learning techniques can analyse highly complex datasets, thereby enhancing fraud detection effectiveness within the financial context (Mishra, 2024). Moreover, techniques such as fuzzy logic have been applied within behavioural finance, offering innovative approaches to risk management (Arora et al., 2023). One study, for example, highlights that the CS-FSM paradigm powered by AI transforms the cybersecurity landscape of the financial sector by enhancing scalability, privacy, data security, risk management, and defence against cyber-attacks (Mishra, 2024). Furthermore, sophisticated AI algorithms and technologies can be leveraged not only for security purposes but also to ensure that ethical values, particularly data privacy and transparency, are rigorously protected.

Various frameworks have been developed based on AI technologies to enhance both security compliance and the ethical utilization of data within banking operations. *"Technological advancements will continuously guide as well as condition the evolution of cybersecurity. The use of AI technology entails growing concerns about compliance with ethics and fundamental rights."* (Iturbe et al., 2023)

In particular, AI-based solutions for credit card fraud detection, online transaction monitoring, and complex financial transactions leverage big data analytics to identify vulnerabilities and predict potentially fraudulent activities through sophisticated algorithms. These technologies not only safeguard customer data during cyber transactions but also support ethical practices in financial contexts by addressing issues related to data privacy and responsible data use. However, further advancements are required to ensure that future developments in AI continue to maximize ethical considerations while addressing emerging security challenges in the financial domain (Nagendiran et al., 2023).

B: Explainable AI (XAI) adoption in banking operations

Explainable AI (XAI) has emerged as a critical approach to address the "black box" nature of conventional AI algorithms, particularly in the banking and financial sectors where transparency and accountability are paramount ethical principles. Interpretable Machine Learning (IML) and XAI methodologies have gained prominence for their ability to provide insights that contribute to ethically responsible AI applications. For instance, a study by Hanif et al., (2021) examines the techniques and challenges associated with explainable AI, emphasizing that to fulfill ethical requirements such as accountability, responsibility, and transparency, XAI offers viable methods for AI system development in the financial sector. Techniques based on XAI including surrogate models, feature attribution, visualization methods, masked data instances, and knowledge graphs are instrumental in enhancing cognitive and computational efficiencies, thereby fostering the development of more responsible AI systems. Nevertheless, this body of research also underscores the importance of incorporating domain-specific expertise from the financial and banking sectors to ensure that explanations provided by AI systems are both effective and comprehensive.

"While there are a number of key features of AI that are essential for its successful adoption in financial services, including transparency (the ability to look within the model and see what data is in it and how it was designed) and robustness (including reliability, repeatability, and scalability), the keystone for safe implementation of AI is explainability." (Adams & Hagras, 2020). In this study, the authors introduce a type-2 fuzzy logic approach to deliver an XAI algorithm that enhances ethical principles, notably transparency and accountability. They propose that explainability is fundamental to unlocking the true benefits of AI adoption in the financial services industry, underscoring accountability as the most important principle of governance and ethics. Their proposed framework aims to ensure that AI outputs are unbiased, accurate, and presented in a manner that is comprehensible to humans.

Another study by Adamyk et al., (2024) highlights that despite the potential benefits of XAI, ML, and other AI technologies in banking and finance, significant barriers remain particularly the high costs of implementation and the challenge of obtaining high-quality data. As it states, *"Only if the bank has a coordinated team of highly qualified AI developers, 'tons' of great data and great processes, the use of complex AI/ML systems will yield excellent results. In the opposite case (if the bank's databases are imperfect, processes are inconsistent and the team of employees is not prepared), the overall risks for the bank are too high"*. The study also emphasizes that the integration of Robotic Process Automation (RPA) with AI technologies can enhance the accuracy and validity of data, thriving ethicality "*Two main use cases for banks using RPA are Know-Your-Customer (KYC) and other due diligence actions for AML and fraud detection. KYC is a rule-based and data-intensive process in any bank. RPA can aggregate client data, evaluate and validate data to accelerate the process and eliminate mistakes.*" Consequently, the authors underscore the importance of using unbiased and unpolluted data in AI systems to support ethically sound and accurate decision-making.

C: Machine Learning (ML) adoption in banking operations

Gaurav et al. (2023) examine how Machine Learning (ML) integrated with Explainable AI (XAI) can enhance transparency and operational efficiency in the digital transformation of the banking sector. By employing XAI techniques, including Sensitivity Analysis (SA) and Layer-wise Relevance Propagation (LRP), the study demonstrates that trust-based AI solutions are achievable. Their findings indicate that the integration of ML with XAI not only optimizes banking processes but also enhances ethical considerations by improving the explainability and interpretability of AI-driven decisions.

Similarly, Lagasio et al. (2022) examine the application of ML in identifying key factors contributing to bank defaults, particularly within the Eurozone financial sector. While their research does not directly address ethical challenges, it underscores the regulatory scrutiny surrounding AI adoption due to potential risks in data handling and automated decision-making. The study highlights the ethical implications of AI-driven financial predictions, emphasizing the necessity of responsible data management, fairness, and transparency in AI models. This perspective reinforces the broader discourse on the ethical use of

AI in banking, particularly in mitigating algorithmic biases and ensuring compliance with financial regulations.

Furthermore, Raman et al. (2024) adopt a mixed-method approach to investigate AI adoption in banking, analysing various AI and ML techniques, including anomaly detection for fraud prevention, logistic regression for credit scoring, and multiple portfolio optimization models. Their study underscores the ethical considerations surrounding AI integration in financial services, particularly in relation to data privacy, algorithmic bias, and decision-making transparency. The authors advocate for financial institutions to prioritize ethical AI practices to enhance consumer trust and regulatory compliance. They argue that addressing biases, fostering accountability, and maintaining transparency are essential for ethical AI adoption, ensuring that technological advancements align with equitable and responsible banking practices (Raman & Tiwari, 2024). Hence, their findings suggest that balancing innovation with ethical responsibility is crucial for fostering sustainable growth in the financial sector.

Guler et al. (2024) emphasize the necessity of interdisciplinary and transdisciplinary research when addressing the ethical and regulatory challenges associated with AI adoption in the banking sector. *"The presence of a cluster dedicated to AI ethics, regulation, and fairness (topics such as ethical AI, legal and regulatory, algorithmic bias, and fairness) highlights the increasing importance of addressing ethical and regulatory challenges associated with AI."*. Their study highlights that a comprehensive approach, integrating technical, ethical, and regulatory perspectives, is essential for ensuring responsible AI implementation in financial institutions. The authors argue that ethical considerations must be embedded within AI-driven business models to mitigate potential risks and enhance regulatory compliance.

In a systematic meta-review of 40 studies on AI and ML applications in banking, Jáuregui-Velarde (2024) identifies several key areas where these technologies significantly contribute. The study discusses customer segmentation using clustering algorithms and underscores the ethical implications of handling vast amounts of financial data. It warns that inadequate data management, potential manipulation, or flawed AI evaluations could undermine consumer confidence and raise ethical concerns related to fairness, privacy, and accountability, stating that, *"the use of AI has revealed numerous ethical problems, such as the incorrect evaluation of credit history, the exchange of false information, and unauthorized operations. Therefore, creating a friendly and ethical AI system is challenging."* Consequently, the review calls for financial institutions to adopt robust AI governance frameworks that prioritize ethical data usage, transparency, and regulatory compliance.

D: Biometric analysis with AI adoptability in banking

A technical study on customer-centric banking through biometric and AI integration, conducted using SLR and case study approach, explores the convergence of biometric identification with AI-driven personalization to enhance customer engagement in banking. *"The combination of biometrics and cryptography also reveals numerous ethical and privacy concerns. The careful handling of privacy,*

permission, and data protection is crucial when dealing with storing, transmitting, and processing biometric data, especially regarding international data protection laws and ethical concerns" (Abdul-Al et al., 2024) through which , the researchers emphasize that while these advancements present substantial opportunities for fostering customer-centric financial services, they also necessitate careful consideration of ethical and regulatory challenges to ensure sustainable growth and differentiation in the banking sector.

Similarly, a study examines the evolution of multi-modal face recognition and its growing role in security biometrics and identity verification through AI. The research discusses the use of Generative Adversarial Networks (GANs), Convolutional Neural Networks (CNNs), and Fuzzy Extractors to enhance the accuracy and security of facial recognition systems. While these technologies strengthen authentication mechanisms, the study underscores the necessity of addressing ethical and privacy considerations to ensure responsible implementation. *"Although the stakeholders were very positive in the way that these technologies could indeed turn their organizations around, issues to do with the breach of data privacy, vulnerability in security, and even compliance with regulation tightness all came out as the very critical considerations."* (Sekar et al., 2024). The authors argue that maintaining a balance between security and ethical obligations is critical, as biometric data is inherently sensitive, and its misuse can lead to significant privacy risks. Therefore, financial institutions must integrate stringent governance frameworks and data protection measures to uphold ethical standards while leveraging biometric AI technologies.

Considering the overall contents of highly technical studies on the use of XAI, ML, and other sophisticated AI technologies in banking operations, nearly all emphasize the critical importance of integrating ethical considerations, regulatory frameworks, and governance structures for responsible AI adoption just in the basic surface level. However, a significant gap remains; none of these studies explicitly detail how ethical AI adoption can be practically achieved. Although many studies acknowledge the substantial vulnerabilities and risks associated with the adoption of certain AI techniques in banking, especially regarding ethical challenges, there is a paucity of research that provides a thorough, in-depth analysis of risk mitigation strategies. Therefore, this analysis reveals a considerable potential for technical experts to develop and present detailed approaches to mitigate ethical concerns and challenges in the context of AI adoption in the banking sector.

5.2.3 AI adoption in banking in legal and regulatory perspective

Ethical challenges in AI adoption within banks and financial institutions are significantly influenced by legal and regulatory perspectives, according to ANT perspective all the actors in the network impact among each other in varied manners. As AI technologies rapidly evolve and expand across diverse industries and contexts, concerns have emerged regarding the development and enforcement of legal and governance frameworks necessary to protect potential victims from both intentional and unintentional misuse of technology. Instances of deliberate illegal use of AI have increased, with reports highlighting how advanced technologies can be exploited as shields for malicious activities, including fraud and other criminal behaviours. A comprehensive study by Rodrigues in 2020, highlights ethical concerns interrelated with legal issues; *"Such issues include: algorithmic transparency, cybersecurity vulnerabilities, unfairness, bias and discrimination, lack of contestability, legal personhood issues, intellectual property issues, adverse effects on workers, privacy and data protection issues, liability for damage and lack of accountability."* (Rodrigues, 2020).

In response to these challenges, efforts are underway to establish robust legal and regulatory frameworks designed to prevent crimes and mitigate the destructive impacts of unethical AI use on individuals, society, and organizations. However, the process of formulating laws and regulations for AI is inherently complex due to significant variations in legal enforcement and political contexts across different regions. Despite these challenges, notable achievements have been made; for example, the European Union has pioneered comprehensive AI-related legislation across its member states, with laws such as the General Data Protection Regulation (GDPR) significantly advancing customer privacy and human rights in the context of technology use.

This study does not aim to provide an in-depth analysis of the legal frameworks and governance regulations related to AI adoption in banks. However, the existing literature reveals that several studies, particularly peer-reviewed conference papers, address critical aspects of the legal perspectives on AI adoption in banking. In the context of ethical use, the challenge of defining "ethical behavior" as described in ethical theories, is closely intertwined with legal standards that regulate acceptable conduct. If it were possible to codify a "concept of ethical use" through law enforcement, AI developers could potentially mitigate ethical challenges by embedding these legal rules into AI systems. The current inability to establish such standardized legal guidelines intensifies the ethical challenges in AI adoption due to the absence of universally accepted rules to govern system behavior. The following sections detail several key legal frameworks that function as governing bodies for AI-related legal concerns. The subcategories are given below:

A: AI adoption in banking with relevance to GDPR

B: AI adoption in banking with relevance to other legal frameworks

A: AI adoption in banking with relevance to GDPR

According to a study by Kamaruddin et al. (2023), the GDPR is regarded as a workable way to manage the advances in data utilization, algorithmic transparency, and accountability of AI technology while simultaneously striking a balance with the fundamental principles of human rights, particularly data protection and privacy, in the face of growing concerns from opposing viewpoints on AI. Currently, the most applicable legal and regulatory framework for protecting privacy and data, defending fundamental human rights, and laying out explicit duties for data controllers is the GDPR. Given that many AI systems rely on big data, the legal issues related to data privacy have been explicitly addressed in the GDPR through key principles; *"These principles include lawfulness, fairness, and transparency, and they need that personal data to be processed lawfully, fairly, and transparently. In addition, the purpose limitation principle stipulates that personal information may only be collected for specific, legitimate, and explicit purposes and not for other incompatible goals."*. Additionally, the framework enshrines individual rights such as the right to be informed, access, rectification, erasure, restriction of processing, data portability, and the right to object which collectively serve to mitigate mass surveillance and enforce ethical standards in data privacy from a legal perspective (Kamaruddin et al., 2023).

A similar study on the application of GDPR to AI ethics underscores several critical components. It emphasizes that obtaining explicit data subject consent is fundamental to lawful data processing. The study delineates core data protection principles, including lawfulness, fairness, transparency, purpose limitation, data minimization, accuracy, storage limitation, security, and accountability, as essential elements of GDPR compliance. The study further integrates ethical principles detailed by virtue ethics, into the discussion, advocating for AI systems to be designed in accordance with the principles of beneficence (promoting societal well-being), non-maleficence (avoiding harm), autonomy (preserving human agency and decision-making), and justice (ensuring fairness and preventing discrimination). Explicability is also identified as a crucial requirement, ensuring that AI systems operate transparently and that individuals can understand the rationale behind automated decisions. *"A human centric AI may give explanations about decisions or estimations. It is clear that there are various approaches to ethics. Robust ethical principles are essential in the future of the rapidly developing AI technology, but not all countries understand ethics in the same way. The current frameworks address the major ethical concerns and make recommendations for governments to manage them, but notable gaps exist."* (Milossi et al., 2021). The study asserts that AI systems should be designed to uphold human dignity and autonomy while maintaining a balance between innovation and ethical responsibility.

Botunac, Parlov and Bosna in 2024 argue that banks must navigate a complex regulatory landscape that includes the EU's AI Act, GDPR, Data Act, and DORA to ensure compliance during the integration of Generative AI technologies. The study underscores the importance of ethical data usage, advocating for the responsible handling of data and the maintenance of transparency and accountability in AI systems. *"The role of international standards such as ISO/IEC*

42001:2023, ISO 31000:2018, ISO/IEC 23894:2023, NIST AI 600-1 and ISO/IEC 23053:2022 is considered to be beneficial in establishing a common framework for managing AI systems, ensuring data integrity and promoting transparency."(Botunac et al., 2024).

B: AI adoption in banking with relevance to other legal frameworks

A study by Tidjon & Khomh, (2023) exploring AI ethics across various global principles and guidelines provides valuable insights into the critical role of legal frameworks in this domain and highlights a significant gap between high-level ethical principles and their practical implementation; *"However, the complexity of ethical AI principles and their context-dependency means that current practical guidance do not capture well their abstraction, leading to gaps"*. Although numerous ethical AI guidelines have been released by governments and organizations, there are notable deficiencies including the lack of practical tools, effective standards, comprehensive training programs, robust corporate governance, and sufficient coverage of ethics. The author identifies several root causes for these gaps; *"The root causes of gaps include a lack of metrics for assessing the practical guidance of ethical principles, poor diversity, and inclusion"*.

The legal frameworks underscore the importance of governance in ensuring that AI systems are accountable, fair, and transparent and the world has been awakened on this issue; *"There are clearly outstanding questions regarding what good AI governance should look like. These questions are currently debated by political institutions across the globe, including the UK, South Korean, the Indian and the Mexican government, as well as the European Commission"* (Cath, 2018). The study advocates for a multidisciplinary approach to AI governance, emphasizing that effective oversight requires the integration of expertise from ethics, law, and technology. It highlights the necessity of adopting a holistic framework that simultaneously considers the ethical, legal, and technical dimensions of AI implementation.

5.2.4 Ethical challenges of AI adoption in explicit perspective

A key observation from the literature review is that, while ethical concerns are frequently mentioned, and merely acknowledge the potential for ethical issues in AI implementation without providing an in-depth examination of specific challenges. In ethical theories viewpoint, the inherent complexity of this domain is the main reason. These works typically lack detailed analyses of how such challenges may manifest, the potential impact on various stakeholders, and the extent to which mitigation strategies are considered both during the development of AI systems and their application within banking operations.

As a result, there exists a considerable gap in the academic literature concerning comprehensive, detailed explorations of ethical issues tied to AI adoption in financial institutions. Nevertheless, this study has identified a limited number of existing publications that offer more explicit and thorough discus-

sions on the ethical implications of AI in banking. The following section outlines key findings from these select studies, which provide valuable insights into the complexities and consequences of ethical challenges that emerge as AI becomes increasingly integrated into banking operations. The subcategories are as follows:

A: Data privacy and security challenges
B: Accountability and liability
C: AI system development challenges
D: Other ethical challenges (fairness, explainability, trust, empathy, human displacement concerns and …etc.)

A: Data privacy and security challenges

This sub-category includes a peer-reviewed journal article and two peer-reviewed conference papers, totalling three studies. Across these studies, data privacy and security emerge as fundamental ethical concerns, given that AI models process vast amounts of data. Numerous scholars have identified data privacy as a critical challenge in AI adoption within the banking sector, with discussions on this issue being present in nearly all reviewed literature, even if only at a superficial level.

A SLR conducted by Nabbosa, (2020) underscores the significant risks associated with data privacy in the digitalized AI era. Author highlights the ethical importance of data collection practices, stating that "*There are other non-monetary benefits such as Trust and Reputation that companies can thrive on if they collect data ethically to develop business solutions, products, services, and processes...*". Additionally, references the International Association of Privacy Professionals (IAPP), which defines data privacy as "*the right to have some control over how your personal information is collected and used*". The author further emphasizes that the increasing accessibility of online data and AI's ability to collect, analyse, and make autonomous decisions heighten the risks to data privacy. Recognizing the severity of these ethical concerns, legal frameworks are being strengthened to impose penalties for violations, as reflected in the observation that "*Financial Penalties arising from Data Privacy violations and other commercial laws are increasing rapidly*". Despite these regulatory efforts, a research gap was identified in this domain for future scholars, "*data privacy and digital information with pressing ethical concerns in today's business environment but we are hopeful that future research will address this research gap*".

Another qualitative study by Nguyen et al., (2024) examines the security and privacy challenges associated with AI and machine learning (ML), asserting that "*The utilization of AI and LLMs in practical applications has encountered a series of hurdles concerning security, privacy, and ethical considerations*". The study provides a technical perspective, detailing how ML-based AI systems can inadvertently expose sensitive information embedded in training data, "*These risks arise from the possibility of sensitive information hidden in the training data being exposed*". Additionally, the study discusses broader ethical concerns such as bias and fairness, transparency and accountability, as well as issues related to authorship and plagiarism. The author underscores the necessity of ethical

frameworks to guide AI development and deployment, advocating for the adoption of established ethical principles "*beneficence, non-maleficence, autonomy, and justice, and provide practical guidelines for ethical AI practices. Developing comprehensive ethical frameworks for AI is necessary to ensure the responsible advancement and development of AI technologies*".

The study by Piotrowski (2023), employed a quantitative survey approach, investigates bank customers' perceptions of data privacy in AI-driven banking operations. The findings reveal widespread public scepticism regarding the use of publicly available data without explicit customer consent. A significant portion of respondents expressed concerns over data privacy breaches, with the study reporting that "*The analysis of the empirical data on personal data processing reveals that a clear majority of respondents have a negative attitude towards sharing data publicly available on social media accounts with banks*". Moreover, the study highlights the ethical implications of AI-based data extraction practices, as a majority of respondents deemed such actions unethical *"72.5% of respondents perceive as unethical the banks' actions of extracting customer information from data in the public domain on their social media accounts*". These findings indicate a growing awareness among customers about AI's utilization of personal data without explicit consent, reinforcing the notion that data privacy remains a pressing ethical challenge. This phenomenon can be further elucidated through the lens of Actor-Network Theory. Within this framework, users are conceptualized as interconnected actors in a vast network, where they are exposed to extensive information regarding ethical issues. Enhanced customer awareness in this context may contribute positively to the mitigation of ethical concerns, such that increased transparency and comprehensive data exposure could help alleviate certain ethical challenges associated with AI adoption in the banking sector.

Overall, these studies collectively emphasize the need for robust ethical and legal frameworks to ensure responsible AI deployment in banking operations, particularly in addressing privacy and security concerns.

B: Accountability and liability

A qualitative study conducted in India by Manoj Kumar et al., (2023) highlights the complexities associated with determining accountability for AI-driven decision-making in the banking sector, as well as the challenges in assigning liability for errors or harm caused by AI systems. The study underscores the intricate nature of responsibility in such cases, stating that "*When AI systems make decisions that have real-world consequences, determining who is accountable for those decisions can be complex*". Despite these challenges, the study emphasizes that customers retain the right to question the accountability of decisions made by AI systems in banking operations. Specifically, it notes that "*If an individual is denied credit due to an algorithmic decision, they may question the accountability of the bank or financial institution. Developing clear policies for explaining algorithmic decisions and providing avenues for review is crucial.*". This highlights the necessity of transparency in AI-driven decision-making and the implementation of structured mechanisms to address customer concerns.

In its conclusion, the study stresses the importance of developing tailored policies and guidelines to ensure a balanced approach that fosters both AI adoption and ethical responsibility. Establishing clear accountability frameworks is imperative to maintaining customer trust and ensuring that AI-driven decisions align with ethical and regulatory standards.

C: Addressing ethics in AI system development for banking

The development of AI systems has been extensively discussed in numerous research papers. However, the consideration of ethical and responsible AI system development remains underexplored in the existing literature. Given the critical role of banks in handling diverse end users across multiple platforms, it is essential to prioritize the development of ethical AI systems tailored to the banking sector.

A study on a conceptual framework for ethical AI development by Chen, (2020) underscores this necessity, stating that "*Authorities and industries are still looking for possibilities to regulate, optimise and harmonise negative issues that have been raised.*". The study further emphasizes the importance of testing and verifying AI systems to ensure their compliance with existing standards and legal frameworks, arguing that such measures can contribute to the advancement of ethical AI development. As part of the study, chatbot services from various sectors, including banking, television companies, municipalities, and airlines, were tested for compliance with ethical and regulatory standards. The results indicated significant shortcomings, as "*The project tested chatbot services among bank, television company, municipality, airline company etc. The result showed none of the chatbots in the test fulfilled the standards of Web Content Accessibility Guidelines (WCAG) 2.0 and 2.1*". The study acknowledges the potential of existing guidelines, policies, and regulations in shaping ethical AI practices, stating that "*With the help of existing guidelines, white papers, policies, regulations, it is possible to develop AI systems that fulfil all requirements, including social and ethical values.*". Furthermore, it highlights the necessity of legal and regulatory interventions, such as the General Data Protection Regulation (GDPR) and other relevant frameworks, to provide a structured approach to ethical and responsible AI development.

Expanding the discussion to a broader perspective, a study by Saleh, (2024) highlights that ethical practices are fundamental for software engineering professionals, another important actor in this ecosystem. The study asserts that ethical considerations must be embedded in software development from multiple perspectives, including the people, process, and product viewpoints. It emphasizes that "*We argue that an ethical software engineer would produce an ethically sound software product by following an ethically sound software development process.*". This principle is not confined solely to AI software development in the banking sector but is applicable to software engineering in general. The study further explains that "*Due to the current advances in the areas of cybersecurity, data analytics and artificial intelligence (AI), it is clear that software engineering ethics is intertwined with data ethics due to the reliance of data-centric secure software systems and AI systems in particular on the proper and ethical gathering, storage, manipulation and dis-*

semination of data.". Given the increasing reliance on AI-driven technologies, the study underscores the pressing need for a well-defined ethical framework to guide the responsible and ethical development of software systems, particularly in the context of AI adoption.

D: Other ethical challenges (explicit)

The discussion of ethics within the context of AI remains largely superficial. Although several studies have addressed digital ethics explicitly, few have examined the intersection of AI and banking in depth. As AI permeates virtually all sectors, including business and social contexts, it is imperative to critically assess the ethical challenges that arise from its adoption in banking operations. The following discussion reviews studies that highlight ethical challenges associated with AI adoption but are not confined to a specific ethical challenge category in this analysis.

The study by Andreasyan et al., (2024) emphasizes the need for comprehensive ethical frameworks, stating, "*It highlights the critical need for establishing ethical guidelines that address bias, transparency, and decision-making. The multifaceted nature of technology underscores the urgency for comprehensive ethical frameworks to navigate the complexities inherent in technological advancements.*" Another study by Ridzuan et al., (2024) focused on AI in the financial sector details ethical responsibilities and regulatory innovations. It explicitly discusses key challenges such as data privacy and security, noting, "*The danger of confidential financial information being accessed illegally. Potential mishandling of client data for unauthorized uses.*" With respect to bias and fairness, the study observes, "*Unintentionally discriminating against particular groups is a possibility with AI systems. Unfair lending and credit score decisions might result from biassed data.*" Challenges in transparency and accountability are also highlighted, with the study stating, "*Decisions made by AI might be complicated and challenging to comprehend. It is unclear who is in charge of AI-driven outcomes.*" This study further proposes solutions for these challenges by discussing existing AI regulations and guidelines and proposing additional strategies for AI governance in financial organizations, such as the EU AI Act and other measures implemented by various countries.

Another study by Rawas, (2024) asserts that AI represents the future of humanity and explicitly details the challenges imposed by AI on human society. It highlights, "*Ethical considerations around data privacy, transparency, and fairness in AI decision-making have become critical issues, prompting the need for robust regulations and ethical guidelines.*" and concludes that "*The future of AI necessitates strong ethical frameworks and norms that value human well-being, fairness, and transparency.*" Furthermore, a study addressing regulatory and ethical issues of AI by Divi et al., (2024) notes that, in addition to commonly discussed ethical challenges, *"Social scoring has emerged as a grave concern from unethical AI usage."*

Additionally, a study by Aitken et al., (2021) explores the value of engaging with the public as a strategy for promoting socially-minded, data-intensive innovation in banking. This comprehensive empirical study declares that "*…awareness of social and political tensions and the power of public movements was felt in the field of data ethics and responsible Artificial Intelligence (AI) where increas-*

ing interest in social justice and socio-technical dimensions of AI and data are said to be underpinning a third wave of ethical AI." The study details various social approaches to addressing data ethics and other ethical challenges that arise with AI adoption in banking operations.

Collectively, these studies reveal in detail that while ethical concerns related to AI adoption are recognized, there remains a significant gap in the explicit, in-depth discussion of these issues within the context of banking.

5.3 Methodological approaches in existing literature

The existing body of literature examined for this systematic literature review was analysed with a particular focus on the methodological approaches adopted in the respective studies. This analysis aimed to identify any potential patterns or relationships between the research methodologies employed and the depth of problematisation of ethical challenges in the context of AI adoption in banking. Furthermore, it sought to assess whether the methodological choices in these studies could either constrain or enhance the potential for conducting a comprehensive and effective survey in this research area.

As the findings indicate, the distribution between methodological approaches is relatively balanced, with 29 studies employing quantitative methods and 28 studies adopting qualitative methodologies. Despite this near equilibrium in research methods, the analysis reveals a noticeable gap: there is a lack of in-depth, critical exploration of the ethical challenges associated with AI adoption in the banking sector, regardless of the methodological approach taken. In addressing RQ3, it can be concluded that the methodological strategies used in the reviewed studies, whether qualitative or quantitative; do not appear to have significantly influenced the depth or nature of the discourse on the key ethical challenges. The findings suggest that, while various methodological tools have been applied, they have not necessarily translated into a more profound or comprehensive analysis of the ethical implications of AI in banking. This highlights the need for more robust, integrative research designs that can better capture the complexities of ethical considerations in this evolving field. In the future research suggestions at the end of the study, more effective methodological approaches have been discussed in detail for deployment.

TABLE 5 Methodologies used in existing literature

Quantitative	Qualitative	SLR	Mixed
(Boustani, 2022) (Baffour Gyau et al., 2024) (Manta et al., 2024) (ALMahadin et al., 2023) (Beckmann & Hark, 2024) (Chernysh et al., 2024) (Durongkadej et al., 2024) (Manser Payne et al., 2021) (Piotrowski & Orzeszko, 2023) (Flavián et al., 2022) (Pfoertsch & Sulaj, 2023) (Akre et al., 2024; Jagadeeswari et al., 2024) (Tiwari et al., 2023) (Balaji, 2024) (Lagasio et al., 2022) (Al Batayneh et al., 2021) (Adams & Hagras, 2020) (Guler et al., 2024) (Jáuregui-Velarde et al., 2024) (Bhatt & Meniya, 2022) (Raman & Tiwari, 2024) (Gaurav & Tiwari, 2023) (Nagendiran et al., 2023) (Abdul-Al et al., 2024) (Piotrowski, 2023) (Andreasyan et al., 2024) (Rawas, 2024)	(Arora et al., 2023) (Mogaji & Nguyen, 2022) (Eskandarany, 2024) (Fundira et al., 2024) (Kanjolia et al., 2024) (Barile et al., 2024) (Kadam et al., 2024) (Johora et al., 2024) (Chitimira et al., 2024) (Dogra & Kumar, 2024) (Iturbe et al., 2023) (Hanif et al., 2021) (Adamyk et al., 2024) (Milossi et al., 2021) (Kamaruddin et al., 2023) (Botunac et al., 2024) (Tidjon & Khomh, 2023) (Cath, 2018) (Evangelista et al., 2023) (Nabbosa, 2020) (Nguyen et al., 2024) (Manoj Kumar et al., 2023) (Chen, 2020) (Saleh, 2024) (Divi et al., 2024) (Aitken et al., 2021)	(Hentzen et al., 2022) (Al-Baity, 2023) (Sadok et al., 2023) (Maurya et al., 2024) (Mehndiratta et al., 2023) (Königstorfer & Thalmann, 2020) (Mathen & Paul, 2025) (Kumar et al., 2021)	(Mohsen et al., 2024) (Sachdeva et al., 2024) (Ramachandran et al., 2024) (Meena et al., 2024) (Gigante & Zago, 2023) (Rohella et al., 2024) (Mishra, 2024) (Sekar et al., 2024) (Ridzuan et al., 2024)

Given below is a graphical representation of above numbers, which depicts the distribution of methodological approaches of literature considered in this study.

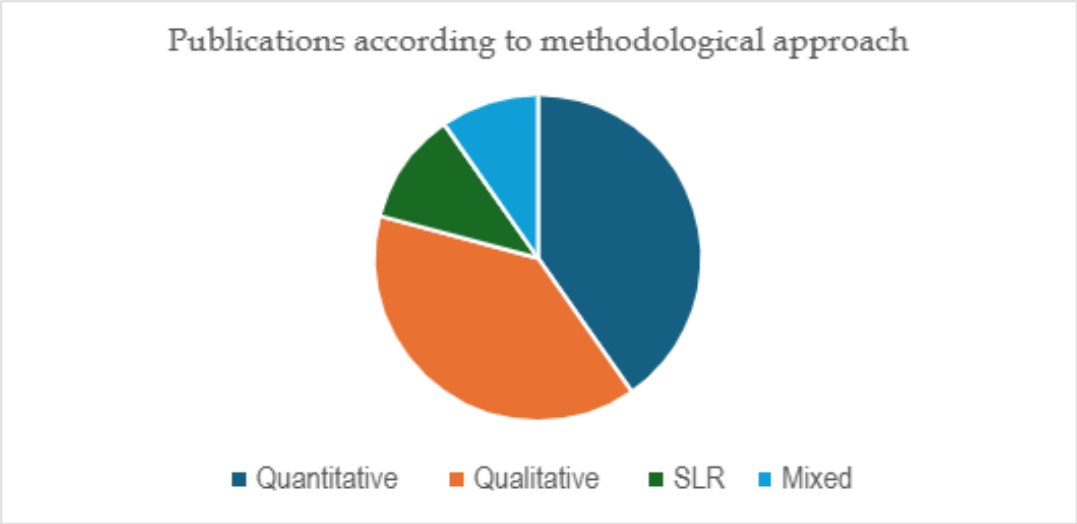


FIGURE 3 The methodological approach used in existing studies

6 DISCUSSION

As the SLR findings demonstrate, the most salient ethical challenge in AI-enabled banking is data privacy and security, an issue that cascades through, and indeed helps to instantiate, other core ethical concerns. From a utilitarian standpoint, violations of privacy generate substantial harm that may outweigh efficiency gains, while a virtue ethics lens highlights the erosion of trust and integrity when customer data is mishandled. These privacy breaches are not simply isolated failures but rather the products of translation processes: data flows (non-human actors) carry embedded interests and assumptions into algorithms, where human actors (developers, bankers, regulators) may inadvertently reinforce biases or weaken accountability. Consequently, bias and discrimination become second-order ethical effects, symptoms of skewed data inscriptions while fairness, transparency, and explainability emerge as necessary mediators to realign network interests around shared ethical norms.

These interlocking challenges directly undermine key adoption constructs from the viewpoint of UTAUT. Privacy and security concerns reduce performance expectancy by casting doubt on AI's reliability and benefit, they increase effort expectancy when users and staff must navigate complex consent or remediation procedures; and they weaken social influence by eroding stakeholder trust in the institution's commitment to responsible innovation. Each of these constructs can be seen as embedded within the socio-technical network where ethical "messages" are translated, sometimes attenuated across actors. The synthesis of findings by SLR had led me to invent a new framework promoting ethical AI adoption in banks which has been detailed in next sub-sections. Then, I will discuss on how stronger interessement process is significant rather than the implicit declaration led to clearer enrolment strategies in technical and explicit perspectives, and more robust mobilization efforts in law and regulation perspective might realign stakeholder incentives and translate ethical principles into sustainable banking practices. At the end of the discussion, future research directions are depicted to continue the studies on this important domain.

6.1 A framework for ethical AI adoption

According to the SLR, the findings from existing research have been extrapolated and converged to form a holistic synthesis of the ethical challenges associated with AI adoption in banking. The discussed categorization revealed interrelated ethical issues, which are then culminated to develop a novel conceptual framework aimed at promoting ethical AI adoption in banking, as depicted in the diagram below.

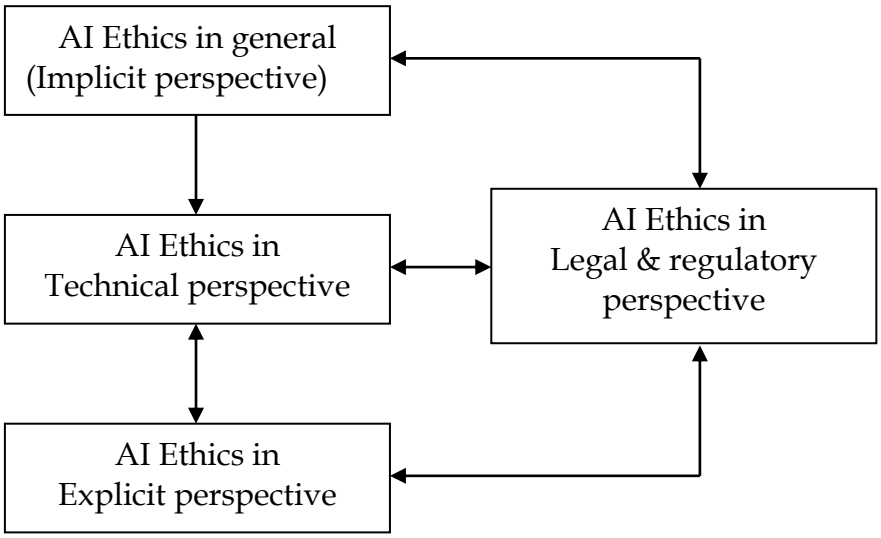


FIGURE 4 Author's conceptual framework for ethical AI adoption in banks

Ethical challenges allied with AI adoption in banking are often addressed implicitly and at a general level in existing literature. However, according to Callon's discussion of the sociology of translation in ANT (Callon, 1986), these concerns are frequently translated into more specialized discourses, such as technical, regulatory, or organizational perspectives. This translation process involves rearticulating general ethical concerns in the language of technical domains, enabling the identification and detailed examination of the technical phenomena that give rise to these concerns, such as algorithmic bias, data privacy vulnerabilities, or decision-making opacity.

This progression from implicit to technical to explicit articulations allows for a more granular understanding of how ethical challenges impact various human and non-human actors within the banking ecosystem. ANT conceptualizes AI as a non-human actor embedded in a dynamic network of other different actors; through its high degree of connectivity and influence, AI reshapes interactions and responsibilities within the system. The ethical implications that emerge from these interactions further influence technology acceptance and adoption behaviours among stakeholders, as explained through the UTAUT.

As illustrated in the accompanying figure above, this framework consists of four interrelated categories: the general (implicit) ethical perspective, the

technical perspective, and the explicit articulation of ethical concerns, all of which are interrelated to legal and regulatory perspectives of AI adoption. These domains do not function in isolation but are dynamically connected through processes of translation and resonance across multiple sub-networks, together constituting the broader socio-technical ecosystem of AI in banking. Central to these interdependencies is the legal and regulatory domain, which systematically shapes and reinforces ethical norms across the other three domains. Academic and industry literature alike emphasize the critical role of regulatory bodies in translating abstract ethical imperatives such as fairness, transparency, and accountability into enforceable standards. However, regulations alone cannot guarantee ethical AI adoption, since ethical norms are inherently context-sensitive and cannot be reduced to fixed rules. Instead, the effective realization of responsible AI requires that legal mandates be embedded within the ongoing translation processes of every network: from developers and system integrators to bank executives and end users. Only through such continuous, network-wide translation can ethical principles resonate with sufficient force to guide transparent and accountable AI deployment in the banking sector.

The conceptual framework I developed, does not exist in the same form within the current body of literature. Rather than being presented as clearly defined, isolated categories, the dimensions I have identified are extrapolated through a synthesis of existing themes and discussions surrounding the ethical challenges of AI adoption in banking. These dimensions are inherently interrelated and overlapping, reflecting the complex and multifaceted nature of ethical considerations in this context.

From the perspective of ANT, ethical concerns interpreted through the theoretical lens of ethical theories, are not confined to individual entities but instead resonate throughout the entire socio-technical network. Within this framework, AI functions as a non-human actor with a particularly strong influence due to its extensive and dynamic interactions with other actors in the network. Its capacity for rapid evolution and deep integration into various operational layers intensively amplifies its impact, often outpacing the adaptive capabilities of traditional regulatory or human actors. Considering the scholarly articles who act as mediators to translate the concept of *ethical AI adoption* among the networks make traceable associations. Those traces are considered while creating accounts or "*texts*" as per Latour, the enormous data transformed as texts, within these scientific networks are important to translate the concepts to other networks as regulations or law enforcements ultimately. The significant differences in translation processes within these networks also impact on the intensity of the message of "ethical use of AI", carried by different actors. (Latour, 2005)

The proposed framework by myself, elucidates how complex ethical challenges emerge within the AI–banking ecosystem by tracing their networked origins and the critical role of problematization in making these challenges visible. By mapping each domain given: implicit ethics, technical considerations,

explicit ethical articulation, and legal or regulatory mandates; the framework locates ethical issues within their specific contexts and proposes collaborative interventions by all network actors. Accordingly, successful ethical AI adoption depends upon the continuous translation and resonance of ethical principles across both human and non-human actors. Absent effective communication and internalization of these principles, the ANT-informed translation process breaks down, allowing ethical imperatives to dissipate and impeding the responsible, sustainable integration of AI in banking. The next section therefore examines the dynamics of translation in greater depth.

6.2 Significance of "translation process" in different networks

The study reveals that "ethical challenges of AI adoption" do not exist as isolated or self-evident phenomena. Rather, these challenges emerge through the dynamic interplay between human actors, technological systems, regulatory structures, and institutional values. The findings indicate that ethical concerns are made visible only through the relationships and interactions that form within socio-technical networks. Most significantly, the academic literature reviewed in this study conveys varying degrees of engagement with ethical issues. The conceptual framework developed above has been impacted by this varied nature of narration by the ethical perspectives. Some articles identify ethics as a critical area requiring further investigation, while many merely acknowledge the presence of ethical concerns without substantive analysis. These articles function as "mentioners" invoking the language of ethics without engaging deeply with its implications. This superficial engagement represents a form of weak translation, as conceptualized by Michel Callon in 1986.

Callon's four moments of translation: problematization, interessement, enrolment, and mobilization underscore the uneven and contested ways in which ethical imperatives are communicated across banking's heterogeneous networks. Regulatory bodies and policymakers, for example, may establish high-level ethical standards, but those norms must be translated through layers of AI developers, software integrators, and hardware vendors before they yield actionable guidance. In my framework, each perspective has their own networks and manipulate the ethical AI adoption in varied ways. Each technical constituency "speaks" its own specialized language, algorithms, code repositories, design specifications, yet all must converge on a shared understanding of responsible AI. The efficacy of this translation, therefore, becomes central to ethical AI adoption. When only every network node internalizes and conveys the same core principles, then the industry can collectively uphold data privacy, fairness, transparency, and accountability.

Considering the studied pool or research articles in this ANT perspective, the problematization is often shallow, with few articles articulating a coherent or detailed framing of the ethical challenges associated with AI in banking. As an example of implicit or general narration; "*As a number of our participants in the*

developing countries expressed certain concerns about the regulatory requirements and awareness of AI in their country, they must consider regulatory advice from other governmental bodies, such as the European Union's ethics guidelines for trustworthy AI, and from tech developers such as Microsoft, who have provided key AI-related ethical principles pertaining to fairness, inclusiveness, reliability and safety, transparency, privacy and security and accountability. In addition, AI systems should involve algorithmic accountability to enable appropriate human direction and control" (Mogaji & Nguyen, 2022). Here varied actors have been mentioned, yet it speaks in superficial manner and depicts lack of genuine intensity towards achieving ethical AI. This text superficially points out the ethical concerns namely, yet the intensity is not strong enough to depict the genuine necessity of in-depth analysis. Hence, underscores a poor translation nature.

Interessement, which involves the process of defining roles and aligning actors around specific ethical positions, is similarly ambiguous. In many cases, the ethical roles assigned to developers, banks, or policymakers are "underdeveloped or assumed" rather than critically analysed. As given in the quoted text above, there are defined roles which are significantly different and requires varied alignment of actors and mediators such as compliance checklists, design protocols and commitment of each actor to fulfil the ethical AI adoption. Yet, the examined literature does not explicitly define the interessement devices and results in poor translation process ultimately.

Enrolment, the alignment of actors into networks with defined responsibilities and relationships is often implicit or fragmented in the literature. Only a small subset of studies provides structured strategies or frameworks that meaningfully link ethical roles to practical implementation. The literature articles are also a part of the network which stand as responsible spokesmen to convey the message of ethical AI adoption. The lack of in-depth information hinders the process of translating the impactful message necessitating consideration for ethical AI strategies adherence by different roles. According to my framework, if explicit perspectives of ethical use such as AI developers' agreeing to integrate explainability modules to decision making, banks adopt governance policies that mandate ethical AI practices are negotiated and formalized, then the translation process will be clearer and more successful.

Consequently, mobilization, the process by which spokespeople or institutions carry ethical messages to broader audiences is uneven. Importantly, the reviewed studies themselves act as actors and intermediaries within a broader network. Through citation, teaching, institutional adoption, and regulatory influence, these studies help shape how ethics in AI adoption is perceived and operationalized. Therefore, the way ethics is handled in academic research carries significant implications: it can shape organizational policy, influence software development norms, and guide regulatory discourse. Articles that explicitly address ethical challenges tend to act as stronger agents of translation, while general or implicit mentions fail to influence practice meaningfully. In this study, there was a significant lack of articles which were rendering in-depth investigation of challenges in ethical AI adoption and ultimately mobiliz-

ing a strong translation of this message to other networks, potentially broader audiences such as regulatory bodies and policy makers.

When implemented in real-world banking operations, Callon's translation framework reveals critical dependencies in ethical AI adoption. For example, a bank's data-privacy protocols such as obtaining explicit KYC consent instantiate regulatory mandates as technical constraints within its systems. However, these constraints will only yield ethical outcomes if they are enrolled by software engineers, product managers, and frontline staff who integrate privacy requirements into actual workflows. When such coherent enrolment is absent, the policy remains a "black-boxed" and intermediary fails to mobilize responsible practice. Moreover, downstream challenges such as algorithmic bias, opaque decision logic, inequitable outcomes, and limited explainability all originate from deficiencies in these initial problematization and interessement phases. Addressing them therefore demands coordinated translation at every network node, ensuring that human actors (developers, compliance officers, business leaders) and non-human actors (data pipelines, AI models, consent interfaces) collectively reinforce privacy, fairness, transparency, and accountability throughout the AI life cycle. Therefore, it is important that the *message of ethical adoption of AI* is effectively translated within the networks to ensure its practical application.

6.3 Future research directions

In light of the identified research gaps, several future research directions can be proposed to advance the understanding of ethical challenges associated with AI adoption in banking. Firstly, the lack of in-depth analysis of ethical challenges in existing literature highlights the need for longitudinal case studies of AI implementation in banks. Such studies would offer empirical insights into how ethical guidelines are applied in real-world contexts and how organizations navigate associated challenges over time. Secondly, this study did not adopt a region-specific analytical approach. Therefore, comparative cross-regional studies, for instance, examining AI ethics in banking across the European Union and Asian countries would be valuable in exploring how cultural, legal, and institutional differences influence ethical challenges and governance frameworks.

Moreover, although this study draws upon ethical theories such as utilitarianism and virtue ethics, there remains a clear gap in operationalizing these theoretical principles within the AI system development lifecycle. Future research could explore how ethical theories can inform practical mechanisms such as bias audits, stakeholder engagement, and transparency protocols throughout system design and deployment. Additionally, with the increasing integration of large language models (LLMs), such as ChatGPT, in financial services, further research is required to examine the ethical implications of generative AI, partic-

ularly in relation to misinformation, informed consent, and transparency in AI-human interactions.

Finally, both qualitative and quantitative studies reviewed in this thesis often lacked depth in capturing the nuanced dimensions of ethical AI adoption. Future research should consider methodological innovations, including mixed-methods approaches, participatory research, and action research frameworks, to better address the complex socio-technical nature of AI ethics in banking. These directions would not only fill existing academic gaps but also contribute to the development of more responsible and context-sensitive AI systems in the financial sector.

Overall, the synthesis of existing literature clearly indicates that ensuring ethical AI adoption in banking requires an integrated approach that addresses data privacy and security as a foundation, while concurrently mitigating related issues of bias, fairness, accountability, transparency, and the preservation of essential human interactions. The new conceptual framework in promoting ethical AI adoption in banks highlights the importance of translations of the message of "ethical AI adoption" within the collaboration of multiple stakeholder actor networks in order to promote sound norms and guidelines for ethical AI development and adoption in banks.

7 CONCLUSION

In conclusion, this study systematically reviewed the ethical challenges associated with AI adoption in the banking sector, synthesizing insights from a diverse range of academic publications. The findings reveal that key ethical concerns, particularly data privacy and security, algorithmic bias, transparency in AI decision-making, and accountability are frequently discussed across the literature. However, these discussions are often superficial, lacking in-depth analysis and practical guidance for mitigating risks. Based on the findings, I have extrapolated four interrelated dimensions which then developed as a new framework promoting a conceptual approach for ethical AI adoption. Theoretical analysis revealed the significance of translation of the message, necessitating of ethical AI adoption in the banking sector.

A notable limitation of this review is the inconsistent depth of analysis in the existing studies, which impedes a fully comprehensive understanding of how these ethical challenges interplay within the banking ecosystem. This underscores a significant gap in the research, particularly regarding the operationalization of ethical principles in practical AI deployments.

Future research should seek to address these gaps through more empirically rigorous studies that provide detailed, context-specific analyses of ethical AI adoption in banking. The novel framework can be referenced to study in-depth how these four dimensions can be manipulated for ethical AI adoption with extended research. In particular, there is a need for research that develops robust metrics for evaluating ethical outcomes and examines the effectiveness of legal and regulatory frameworks in guiding responsible AI practices. Advances in these areas will not only enrich academic discourse but also support the development of ethically sound and accountable AI systems within the financial sector.

REFERENCES

A Proposal for the Dartmouth Summer Research Project on Artificial Intelligence, August 31, 1955 | AI Magazine. (n.d.). Retrieved 18 February 2025, from https://ojs.aaai.org/aimagazine/index.php/aimagazine/article/view/1904

Abakarim, Y., Lahby, M., & Attioui, A. (2018). Towards An Efficient Real-time Approach To Loan Credit Approval Using Deep Learning. *2018 9th International Symposium on Signal, Image, Video and Communications (ISIVC)*, 306–313. https://doi.org/10.1109/ISIVC.2018.8709173

Abdul-Al, M., Kumi Kyeremeh, G., Qahwaji, R., Ali, N. T., & Abd-Alhameed, R. A. (2024). The Evolution of Biometric Authentication: A Deep Dive Into Multi-Modal Facial Recognition: A Review Case Study. *IEEE Access*, *12*, 179010–179038. https://doi.org/10.1109/ACCESS.2024.3486552

Acquisti, A., Brandimarte, L., & Loewenstein, G. (2020). Secrets and Likes: The Drive for Privacy and the Difficulty of Achieving It in the Digital Age. *Journal of Consumer Psychology*, *30*(4), 736–758. https://doi.org/10.1002/jcpy.1191

Adams, J., & Hagras, H. (2020). A Type-2 Fuzzy Logic Approach to Explainable AI for regulatory compliance, fair customer outcomes and market stability in the Global Financial Sector. *2020 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, 1–8. https://doi.org/10.1109/FUZZ48607.2020.9177542

Adamyk, B., Benson, V., Adamyk, O., Fraczek, B., & Chinnaswamy, A. (2024). Barriers to Trust in AI: A Study of the Explainability Technologies Adoption in Banks. *2024 14th International Conference on Advanced Computer Information Technologies (ACIT)*, 230–234. https://doi.org/10.1109/ACIT62333.2024.10712606

Adeyelu, O., Ugochukwu, C., & Shonibare, M. (2024). ETHICAL IMPLICATIONS OF AI IN FINANCIAL DECISION – MAKING: A REVIEW WITH REAL WORLD APPLICATIONS. *International Journal of Applied Research in Social Sciences*, *6*, 608–630. https://doi.org/10.51594/ijarss.v6i4.1033

Aitken, M., Ng, M., Horsfall, D., Coopamootoo, K. P. L., Van Moorsel, A., & Elliott, K. (2021). In pursuit of socially-minded data-intensive innovation in banking: A focus group study of public expectations of digital innovation in banking. *Technology in Society*, *66*, 101666. https://doi.org/10.1016/j.techsoc.2021.101666

Akre, P. D., Pacharaney, U., & Siraskar, W. (2024). Smart Finance: An overview of Artificial Intelligence Integration in Fintech. *2024 2nd DMIHER International Conference on Artificial Intelligence in Healthcare, Education and Industry (IDICAIEI)*, 1–6. https://doi.org/10.1109/IDICAIEI61867.2024.10842818

Al Batayneh, A. A., Qasaimeh, M., & Al-Qassas, R. S. (2021). A Scoring System for Information Security Governance Framework Using Deep Learning Algorithms: A Case Study on the Banking Sector. *Journal of Data and Information Quality*, *13*(2), 1–34. https://doi.org/10.1145/3418172

Al-Baity, H. H. (2023). The Artificial Intelligence Revolution in Digital Finance in Saudi Arabia: A Comprehensive Review and Proposed Framework. *Sustainability*, *15*(18), 13725. https://doi.org/10.3390/su151813725

ALMahadin, G., Sawant, P. D., Ali, S., Anjum, A., Ibrahim, S. K., & Naser, S. J. (2023). Enabling Smart Banking AI and IoT: Challenges and Opportunities. *2023 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, 1–6. https://doi.org/10.1109/ICSES60034.2023.10465372

Andreasyan, N., Buson, D., Rüst, J., Portmann, E., & Terán, L. (2024). Theoretical Framework of Digital Ethics Concerns for Public Services: Electronic Voting Use Case. *2024 Tenth International Conference on eDemocracy & eGovernment (ICEDEG)*, 1–9. https://doi.org/10.1109/ICEDEG61611.2024.10702098

Arora, A., Gupta, S., Devi, C., & Walia, N. (2023). Customer experiences in the era of artificial intelligence (AI) in context to FinTech: A fuzzy AHP approach. *Benchmarking: An International Journal*, *30*(10), 4342–4369. https://doi.org/10.1108/BIJ-10-2021-0621

Baffour Gyau, E., Appiah, M., Gyamfi, B. A., Achie, T., & Naeem, M. A. (2024). Transforming banking: Examining the role of AI technology innovation in boosting banks financial performance. *International Review of Financial Analysis*, *96*, 103700. https://doi.org/10.1016/j.irfa.2024.103700

Balaji, K. (2024). Artificial Intelligence for Enhanced Anti-Money Laundering and Asset Recovery: A New Frontier in Financial Crime Prevention. *2024 Second International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI)*, 1010–1016. https://doi.org/10.1109/ICoICI62503.2024.10696100

Barile, D., Secundo, G., & Bussoli, C. (2024). Exploring artificial intelligence robo-advisor in banking industry: A platform model. *Management Decision*, *ahead-of-print*(ahead-of-print). https://doi.org/10.1108/MD-08-2023-1324

Baumane-Vitolina, I., Cals, I., & Sumilo, E. (2016). Is Ethics Rational? Teleological, Deontological and Virtue Ethics Theories Reconciled in the Context of Traditional Economic Decision Making. *Procedia Economics and Finance*, *39*, 108–114. https://doi.org/10.1016/S2212-5671(16)30249-0

Beckmann, L., & Hark, P. F. (2024). ChatGPT and the banking business: Insights from the US stock market on potential implications for banks. *Finance Research Letters*, *63*, 105237. https://doi.org/10.1016/j.frl.2024.105237

Belenguer, L. (2022). AI bias: Exploring discriminatory algorithmic decision-making models and the application of possible machine-centric solutions adapted from the pharmaceutical industry. *AI and Ethics*, *2*(4), 771–787. https://doi.org/10.1007/s43681-022-00138-8

Bencherki, N. (2017). Actor–Network Theory. In *The International Encyclopedia of Organizational Communication* (pp. 1–13). John Wiley & Sons, Ltd. https://doi.org/10.1002/9781118955567.wbieoc002

Bentham, J. (1994). *An Introduction to the Principles of Morals and Legislation*.

Bhatt, D. H., & Meniya, A. (2022). A Review on Machine Learning Methods for Credit Card Fraud Classification. *2022 Second International Conference on Artificial Intelligence and Smart Energy (ICAIS)*, 312–318. https://doi.org/10.1109/ICAIS53314.2022.9743014

Bhattacharya, J. (2024, November 23). The Impact of AI in Banking: Key Statistics and Trends. *SEO Sandwitch*. https://seosandwitch.com/ai-in-banking-stats/

Boell, S. K., & Cecez-Kecmanovic, D. (2014). A Hermeneutic Approach for Conducting Literature Reviews and Literature Searches. *Communications of the Association for Information Systems*, *34*. https://doi.org/10.17705/1CAIS.03412

Boell, S. K., & Cecez-Kecmanovic, D. (2015). On being 'Systematic' in Literature Reviews in IS. *Journal of Information Technology*, *30*(2), 161–173. https://doi.org/10.1057/jit.2014.26

Botunac, I., Parlov, N., & Bosna, J. (2024). Opportunities of Gen AI in the Banking Industry with regards to the AI Act, GDPR, Data Act and DORA. *2024 13th Mediterranean Conference on Embedded Computing (MECO)*, 1–6. https://doi.org/10.1109/MECO62516.2024.10577936

Boustani, N. M. (2022). Artificial intelligence impact on banks clients and employees in an Asian developing country. *Journal of Asia Business Studies*, *16*(2), 267–278. https://doi.org/10.1108/JABS-09-2020-0376

Callon, M. (1986). Some elements of a sociology of translation: Domestication of the scallops and the fishermen of St Brieuc Bay. *The Sociological Review*. https://www.academia.edu/75607686/Some_elements_of_a_sociology_of_translation_domestication_of_the_scallops_and_the_fishermen_of_St_Brieuc_Bay

Callon, M., & Law, J. (1997). After the Individual in Society: Lessons on Collectivity from Science, Technology and Society. *The Canadian Journal of Sociology / Cahiers Canadiens de Sociologie*, *22*(2), 165–182. https://doi.org/10.2307/3341747

Carson et al. (2020). *AI in banking: Can banks meet the challenge? | McKinsey*. https://www.mckinsey.com/industries/financial-services/our-insights/ai-bank-of-the-future-can-banks-meet-the-ai-challenge

Cath, C. (2018). Governing artificial intelligence: Ethical, legal and technical opportunities and challenges. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, *376*(2133), 20180080. https://doi.org/10.1098/rsta.2018.0080

Chen, L. Y.-H. (2020). A conceptual framework for AI system development and sustainable social equality. *2020 IEEE / ITU International Conference on Artificial Intelligence for Good (AI4G)*, 101–106. https://doi.org/10.1109/AI4G50087.2020.9310984

Chernysh, O., Smishko, O., Koverninska, Y., Prokopenko, M., & Pistunov, I. (2024). The Role of Artificial Intelligence in Financial Analysis and Forecasting: Using Data and Algorithms. *Data and Metadata*, *3*. https://doi.org/10.56294/dm2024.224

Chitimira, H., Torerai, E., & Jana, L. (2024). Leveraging Artificial Intelligence to Combat Money Laundering and Related Crimes in the South African Banking Sector. *Potchefstroom Electronic Law Journal*, *27*. https://doi.org/10.17159/1727-3781/2024/v27i0a18024

Chopra, P. (2024). Ethical Implications of AI in Financial Services: Bias, Transparency, and Accountability. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, *10*(5), Article 5. https://doi.org/10.32628/CSEIT241051017

Collins, C., Dennehy, D., Conboy, K., & Mikalef, P. (2021). Artificial intelligence in information systems research: A systematic literature review and research agenda. *International Journal of Information Management*, *60*, 102383. https://doi.org/10.1016/j.ijinfomgt.2021.102383

Deloitte Insights. (2022). *Unleashing the power of machine learning models in banking through explainable artificial intelligence (XAI)*. Deloitte Insights. https://www2.deloitte.com/us/en/insights/industry/financial-services/explainable-ai-in-banking.html

Divi, S., Sahoo, M. K., & Minampati, V. R. (2024). Digital Transaction for New Economic Transformation of India: Regulatory and Ethical Issues in the Use of AI. *Journal of Ecohumanism*, *3*(7). https://doi.org/10.62754/joe.v3i7.4377

Dogra, D., & Kumar, S. (2024). Utilizing Robot Kinematics and Dynamics for Enhanced Debt Recovery in the Banking Sector. *2024 5th International Conference on Image Processing and Capsule Networks (ICIPCN)*, 655–659. https://doi.org/10.1109/ICIPCN63822.2024.00113

Doumpos, M., Zopounidis, C., Gounopoulos, D., Platanakis, E., & Zhang, W. (2023). Operational research and artificial intelligence methods in banking. *European Journal of Operational Research*, *306*(1), 1–16. https://doi.org/10.1016/j.ejor.2022.04.027

Durongkadej, I., Hu, W., & Wang, H. E. (2024). How artificial intelligence incidents affect banks and financial services firms? A study of five firms. *Finance Research Letters*, *70*, 106279. https://doi.org/10.1016/j.frl.2024.106279

Eskandarany, A. (2024). Adoption of artificial intelligence and machine learning in banking systems: A qualitative survey of board of directors. *Frontiers in Artificial Intelligence*, *7*, 1440051. https://doi.org/10.3389/frai.2024.1440051

Evangelista, I. R. S., Bandala, A. A., Culaba, A. B., Concepcion, R. S., & Dadios, E. P. (2023). Technology Adoption: A Policy-Centric Analysis of the Philippines' Preparedness to Embrace Conversational AI. *2023 IEEE 15th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, 1–6. https://doi.org/10.1109/HNICEM60674.2023.10589020

Fares, O. H., Butt, I., & Lee, S. H. M. (2023). Utilization of artificial intelligence in the banking sector: A systematic literature review. *Journal of Financial Services Marketing*, *28*(4), 835–852. https://doi.org/10.1057/s41264-022-00176-7

Flavián, C., Pérez-Rueda, A., Belanche, D., & Casaló, L. V. (2022). Intention to use analytical artificial intelligence (AI) in services – the effect of technology readiness and awareness. *Journal of Service Management*, *33*(2), 293–320. https://doi.org/10.1108/JOSM-10-2020-0378

French, A. M., & Shim, J. P. (2024). From Artificial Intelligence to Augmented Intelligence: A Shift in Perspective, Application, and Conceptualization of AI. *Information Systems Frontiers*. https://doi.org/10.1007/s10796-024-10562-2

Fundira, M., Edoun, E. I., & Pradhan, A. (2024). Evaluating end-users' digital competencies and ethical perceptions of AI systems in the context of sustainable digital banking. *Sustainable Development*, *32*(5), 4866–4878. https://doi.org/10.1002/sd.2945

Galazova, S. S., & Magomaeva, L. R. (2019). The Transformation of Traditional Banking Activity in Digital. *International Journal of Economics and Business Administration*, *VII*(Special Issue 2), 41–51. https://doi.org/10.35808/ijeba/369

Gaurav, D., & Tiwari, S. (2023). Interpretability Vs Explainability: The Black Box of Machine Learning. *2023 International Conference on Computer Science, Information Technology and Engineering (ICCoSITE)*, 523–528. https://doi.org/10.1109/ICCoSITE57641.2023.10127717

Gigante, G., & Zago, A. (2023). DARQ technologies in the financial sector: Artificial intelligence applications in personalized banking. *Qualitative Research in Financial Markets*, *15*(1), 29–57. https://doi.org/10.1108/QRFM-02-2021-0025

Guler, N., Kirshner, S. N., & Vidgen, R. (2024). A literature review of artificial intelligence research in business and management using machine learning and ChatGPT. *Data and Information Management*, *8*(3), 100076. https://doi.org/10.1016/j.dim.2024.100076

Hale, K. (2025). *A.I. Bias Caused 80% Of Black Mortgage Applicants To Be Denied*. Forbes. https://www.forbes.com/sites/korihale/2021/09/02/ai-bias-caused-80-of-black-mortgage-applicants-to-be-denied/

Hanif, A., Zhang, X., & Wood, S. (2021). A Survey on Explainable Artificial Intelligence Techniques and Challenges. *2021 IEEE 25th International Enterprise Distributed Object Computing Workshop (EDOCW)*, 81–89. https://doi.org/10.1109/EDOCW52865.2021.00036

Hentzen, J. K., Hoffmann, A., Dolan, R., & Pala, E. (2022). Artificial intelligence in customer-facing financial services: A systematic literature review and agenda for future research. *International Journal of Bank Marketing*, *40*(6), 1299–1336. https://doi.org/10.1108/IJBM-09-2021-0417

Iturbe, E., Rios, E., Rego, A., & Toledo, N. (2023). Artificial Intelligence for next generation cybersecurity: The AI4CYBER framework. *Proceedings of the*

18th International Conference on Availability, Reliability and Security, 1–8. https://doi.org/10.1145/3600160.3605051

Jagadeeswari, T., Aggarwal, S., Pediredla, R., Chandra, S., Gupta, M., & Katta, S. (2024). AI-Driven Quantitative Evaluation of the Impact of Financial Regulations and HR Marketing on Bank Performance A Cross-Country Analysis. *2024 International Conference on Intelligent Computing and Emerging Communication Technologies (ICEC)*, 1–6. https://doi.org/10.1109/ICEC59683.2024.10837266

Jáuregui-Velarde, R., Andrade-Arenas, L., Molina-Velarde, P., & Yactayo-Arias, C. (2024). Financial revolution: A systemic analysis of artificial intelligence and machine learning in the banking sector. *International Journal of Electrical and Computer Engineering (IJECE)*, *14*(1), 1079. https://doi.org/10.11591/ijece.v14i1.pp1079-1090

Johora, F. T., Hasan, R., Farabi, S. F., Alam, M. Z., Sarkar, M. I., & Al Mahmud, M. A. (2024). AI Advances: Enhancing Banking Security with Fraud Detection. *2024 First International Conference on Technological Innovations and Advance Computing (TIACOMP)*, 289–294. https://doi.org/10.1109/TIACOMP64125.2024.00055

Kadam, S. K., Anawade, P., & Sharma, D. S. (2024). AI-Driven Customer Relationship Management a Game Changer for Indian Banks. *2024 5th International Conference on Data Intelligence and Cognitive Informatics (ICDICI)*, 1503–1506. https://doi.org/10.1109/ICDICI62993.2024.10810901

Kamaruddin, S., Mohammad, A. M., Saufi, N. N. M., Rosli, W. R. W., Othman, M. B., & Hamin, Z. (2023). Compliance to GDPR Data Protection and Privacy in Artificial Intelligence Technology: Legal and Ethical Ramifications in Malaysia. *2023 International Conference on Disruptive Technologies (ICDT)*, 284–288. https://doi.org/10.1109/ICDT57929.2023.10150615

Kamleitner, B., & Mitchell, V. (2019). Your Data Is My Data: A Framework for Addressing Interdependent Privacy Infringements. *Journal of Public Policy & Marketing*, *38*(4), 433–450. https://doi.org/10.1177/0743915619858924

Kanjolia, A., Tiwari, A., Kumar, B., & Mehra, P. S. (2024). Harnessing ChatGPT in Financial Services: Opportunities and Challenges. *2024 1st International Conference on Advanced Computing and Emerging Technologies (ACET)*, 1–6. https://doi.org/10.1109/ACET61898.2024.10730216

Königstorfer, F., & Thalmann, S. (2020). Applications of Artificial Intelligence in commercial banks – A research agenda for behavioral finance. *Journal of Behavioral and Experimental Finance*, *27*, 100352. https://doi.org/10.1016/j.jbef.2020.100352

Kumar, A., Sharma, S., & Mahdavi, M. (2021). Machine Learning (ML) Technologies for Digital Credit Scoring in Rural Finance: A Literature Review. *Risks*, *9*(11), 192. https://doi.org/10.3390/risks9110192

Kurshan, E., Chen, J., Storchan, V., & Shen, H. (2022). On the current and emerging challenges of developing fair and ethical AI solutions in

financial services. *Proceedings of the Second ACM International Conference on AI in Finance*, 1–8. https://doi.org/10.1145/3490354.3494408

Lagasio, V., Pampurini, F., Pezzola, A., & Quaranta, A. G. (2022). Assessing bank default determinants via machine learning. *Information Sciences*, *618*, 87–97. https://doi.org/10.1016/j.ins.2022.10.128

Latour, B. (with Internet Archive). (2005). *Reassembling the social: An introduction to actor-network-theory*. Oxford ; New York : Oxford University Press. http://archive.org/details/reassemblingsoci0000lato

Lazo, M., & Ebardo, R. (2023). Artificial Intelligence Adoption in the Banking Industry: Current State and Future Prospect. *Journal of Innovation Management*, *11*. https://doi.org/10.24840/2183-0606_011.003_0003

Manoj Kumar, K., Madhu, M., Pratyaksha, B., Sushmita, S., & Javed, G. S. (2023). Ethical AI Conundrum: Accountability and Liability of AI decision making. *2023 IEEE Technology & Engineering Management Conference - Asia Pacific (TEMSCON-ASPAC)*, 1–6. https://doi.org/10.1109/TEMSCON-ASPAC59527.2023.10531445

Manser Payne, E. H., Peltier, J., & Barger, V. A. (2021). Enhancing the value co-creation process: Artificial intelligence and mobile banking service platforms. *Journal of Research in Interactive Marketing*, *15*(1), 68–85. https://doi.org/10.1108/JRIM-10-2020-0214

Manta, A. G., Bădîrcea, R. M., Doran, N. M., Badareu, G., Gherțescu, C., & Popescu, J. (2024). Industry 4.0 Transformation: Analysing the Impact of Artificial Intelligence on the Banking Sector through Bibliometric Trends. *Electronics*, *13*(9), 1693. https://doi.org/10.3390/electronics13091693

Martin, K. D., & Zimmermann, J. (2024). Artificial intelligence and its implications for data privacy. *Current Opinion in Psychology*, *58*, 101829. https://doi.org/10.1016/j.copsyc.2024.101829

Mathen, M. P., & Paul, A. (2025). Toward an evolving framework for responsible AI for credit scoring in the banking industry. *Journal of Information, Communication and Ethics in Society*, *23*(1), 148–163. https://doi.org/10.1108/JICES-08-2024-0122

Maurya, S., Verma, R., Khilnani, L., Bhakuni, A. S., Kumar, M., & Rakesh, N. (2024). Effect of AI on the Financial Sector: Risk Control, Investment Decision-making, and Business Outcome. *2024 11th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)*, 1–7. https://doi.org/10.1109/ICRITO61523.2024.10522410

McCarthy, J. (1959). *PROGRAMS WITH COMMON SENSE*.

Meena, R., Mishra, A. K., & Raut, R. K. (2024). Strategic insights: Mapping the terrain of artificial intelligence (AI) in banking through mixed method approach. *VINE Journal of Information and Knowledge Management Systems*, *ahead-of-print*(ahead-of-print). https://doi.org/10.1108/VJIKMS-01-2024-0028

Mehndiratta, N., Arora, G., & Bathla, R. (2023). The use of Artificial Intelligence in the Banking Industry. *2023 International Conference on Recent Advances*

in Electrical, Electronics & Digital Healthcare Technologies (REEDCON), 588–591. https://doi.org/10.1109/REEDCON57544.2023.10150681

Milossi, M., Alexandropoulou-Egyptiadou, E., & Psannis, K. E. (2021). AI Ethics: Algorithmic Determinism or Self-Determination? The GPDR Approach. *IEEE Access*, *9*, 58455–58466. IEEE Access. https://doi.org/10.1109/ACCESS.2021.3072782

Mishra, S. (2024). The impact of AI-based cyber security on the banking and financial sectors. *Journal of Cybersecurity and Information Management*, *14*(1), 08–19. https://doi.org/10.54216/JCIM.140101

Mogaji, E., & Nguyen, N. P. (2022). Managers' understanding of artificial intelligence in relation to marketing financial services: Insights from a cross-country study. *International Journal of Bank Marketing*, *40*(6), 1272–1298. https://doi.org/10.1108/IJBM-09-2021-0440

Mohsen, S. E., Hamdan, A., & Shoaib, H. M. (2024). Digital transformation and integration of artificial intelligence in financial institutions. *Journal of Financial Reporting and Accounting*, *ahead-of-print*(ahead-of-print). https://doi.org/10.1108/JFRA-09-2023-0544

Mossavar-Rahmani, F., & Zohuri, B. (2023). The Transformative Impact of AI on Financial Institutions, with a Focus on Banking. *Journal of Engineering and Applied Sciences Technology*, *5*, 1–6. https://doi.org/10.47363/JEAST/2023(5)192

Mudrack, P. E., & Mason, E. S. (2013). Ethical Judgments: What Do We Know, Where Do We Go? *Journal of Business Ethics*, *115*(3), 575–597. https://doi.org/10.1007/s10551-012-1426-z

Nabbosa, V. (2020). Me Too: Value Creation by Digitalization and Data Privacy. *Proceedings of the 2020 4th International Conference on E-Education, E-Business and E-Technology*, 20–24. https://doi.org/10.1145/3404649.3404650

Nabbosa, V., & Kaar, C. (2020). Societal and Ethical Issues of Digitalization. *Proceedings of the 2020 International Conference on Big Data in Management*, 118–124. https://doi.org/10.1145/3437075.3437093

Nagendiran, S., Renugadevi, R., Kumar, R. P. A., Sasirekha, K., & Harini, R. (2023). Secure Sensitive Information on IoT using Machine Learning. *2023 International Conference on Sustainable Communication Networks and Application (ICSCNA)*, 360–366. https://doi.org/10.1109/ICSCNA58489.2023.10370157

NBER. (2019). *Minority Borrowers Pay More, Even under Algorithmic Lending*. NBER. https://www.nber.org/digest/oct19/minority-borrowers-pay-more-even-under-algorithmic-lending

Nguyen, T. T., Vu, H. T. T., & Nguyen, H. N. (2024). Security, Privacy, and Ethical Challenges of Artificial Intelligence in Large Language Model Scope: A Comprehensive Survey. *2024 1st International Conference On Cryptography And Information Security (VCRIS)*, 1–6. https://doi.org/10.1109/VCRIS63677.2024.10813382

Novelli, C., Taddeo, M., & Floridi, L. (2024). Accountability in artificial intelligence: What it is and how it works. *AI & SOCIETY*, *39*(4), 1871–1882. https://doi.org/10.1007/s00146-023-01635-y

Peters, U., & Carman, M. (2024). Cultural Bias in Explainable AI Research: A Systematic Analysis. *J. Artif. Int. Res.*, *79*. https://doi.org/10.1613/jair.1.14888

Petersen, K., Vakkalanka, S., & Kuzniarz, L. (2015). Guidelines for conducting systematic mapping studies in software engineering: An update. *Information and Software Technology*, *64*, 1–18. https://doi.org/10.1016/j.infsof.2015.03.007

Pfoertsch, W., & Sulaj, K. (2023). Integrating Artificial Intelligence with Customer Experience in Banking: An Empirical Study on how Chatbots and Virtual Assistants Enhance Empathy. *2023 International Conference on Computing, Networking, Telecommunications & Engineering Sciences Applications (CoNTESA)*, 69–74. https://doi.org/10.1109/CoNTESA61248.2023.10384979

Piotrowski, D. (2023). Privacy frontiers in customers' relations with banks. *Economics and Business Review*, *9*(1), 119–141. https://doi.org/10.18559/ebr.2023.1.5

Piotrowski, D., & Orzeszko, W. (2023). Artificial intelligence and customers' intention to use robo-advisory in banking services. *Equilibrium. Quarterly Journal of Economics and Economic Policy*, *18*(4), 967–1007. https://doi.org/10.24136/eq.2023.031

Polireddi, N. S. A. (2024). An effective role of artificial intelligence and machine learning in banking sector. *Measurement: Sensors*, *33*, 101135. https://doi.org/10.1016/j.measen.2024.101135

PRISMA statement. (2020). PRISMA 2020. https://www.prisma-statement.org

Qureshi, N. I., Choudhuri, S. S., Nagamani, Y., Varma, R. A., & Shah, R. (2024). Ethical Considerations of AI in Financial Services: Privacy, Bias, and Algorithmic Transparency. *2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS)*, *1*, 1–6. https://doi.org/10.1109/ICKECS61492.2024.10616483

Rahman, M., Ming, T. H., Baigh, T. A., & Sarker, M. (2023). Adoption of artificial intelligence in banking services: An empirical analysis. *International Journal of Emerging Markets*, *18*(10), 4270–4300. https://doi.org/10.1108/IJOEM-06-2020-0724

Ramachandran, K. K., Takhar, S., Jha, M. K., Patel, J. D., Randhawa, N., & Lourens, M. (2024). Revolutionising Industries and Empowering Human Potential with Artificial Intelligence Tools and Applications. *2024 International Conference on Trends in Quantum Computing and Emerging Business Technologies*, 1–6. https://doi.org/10.1109/TQCEBT59414.2024.10545285

Raman, R., & Tiwari, P. (2024). Finance's AI Revolution: Transforming Banking and Investments for Tomorrow. *2024 International Conference on Emerging Systems and Intelligent Computing (ESIC)*, 478–483. https://doi.org/10.1109/ESIC60604.2024.10481645

Rawas, S. (2024). AI: The future of humanity. *Discover Artificial Intelligence*, *4*(1), 25. https://doi.org/10.1007/s44163-024-00118-3

Ridzuan, N. N., Masri, M., Anshari, M., Fitriyani, N. L., & Syafrudin, M. (2024). AI in the Financial Sector: The Line between Innovation, Regulation and Ethical Responsibility. *Information*, *15*(8), 432. https://doi.org/10.3390/info15080432

Rodrigues, R. (2020). Legal and human rights issues of AI: Gaps, challenges and vulnerabilities. *Journal of Responsible Technology*, *4*, 100005. https://doi.org/10.1016/j.jrt.2020.100005

Rohella, P., Temara, S., Samanthapudi, S. V., Gupta, K., & Suganyadevi, K. (2024). Investigating the Role of AI in Personalized Consumer Banking Experiences. *2024 International Conference on E-Mobility, Power Control and Smart Systems (ICEMPS)*, 1–6. https://doi.org/10.1109/ICEMPS60684.2024.10559374

Ryyan. (2025). *Rayyan: AI-Powered Systematic Review Management Platform*. https://www.rayyan.ai/

Sachdeva, C., Gangwar, V. P., Grover, V., & Gochhait, S. (2024). Cognitive Dissonance in Banking Employees: Exploring Factors Amid the Artificial Intelligence Revolution. *2024 ASU International Conference in Emerging Technologies for Sustainability and Intelligent Systems (ICETSIS)*, 1731–1735. https://doi.org/10.1109/ICETSIS61505.2024.10459558

Sadok, H., Mahboub, H., Chaibi, H., Saadane, R., & Wahbi, M. (2023). Applications of Artificial Intelligence in Finance: Prospects, Limits and Risks. *2023 International Conference on Digital Age & Technological Advances for Sustainable Development (ICDATA)*, 145–149. https://doi.org/10.1109/ICDATA58816.2023.00034

Saleh, K. (2024). Extended Principlism Approach to Software Engineering Ethics. *2024 4th International Conference on Information Communication and Software Engineering (ICICSE)*, 51–54. https://doi.org/10.1109/ICICSE61805.2024.10625674

Sekar, S., Dutta, C., R, S., Mohammed, N. Q., Al-Khuzaie, M. Y., & Almulla, A. A. (2024). A Well Design of Customer Centric Based Financing Service through Biometric Based Veritable System using Personalized AI Integrated System. *2024 4th International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)*, 1109–1113. https://doi.org/10.1109/ICACITE60783.2024.10616506

Siemens, G., Marmolejo-Ramos, F., Gabriel, F., Medeiros, K., Marrone, R., Joksimovic, S., & De Laat, M. (2022). Human and artificial cognition. *Computers and Education: Artificial Intelligence*, *3*, 100107. https://doi.org/10.1016/j.caeai.2022.100107

Tenbrunsel, A. E., Diekmann, K. A., Wade-Benzoni, K. A., & Bazerman, M. H. (2010). The ethical mirage: A temporal explanation as to why we are not as ethical as we think we are. *Research in Organizational Behavior*, *30*, 153–173. https://doi.org/10.1016/j.riob.2010.08.004

Tidjon, L. N., & Khomh, F. (2023). The Different Faces of AI Ethics Across the World: A Principle-to-Practice Gap Analysis. *IEEE Transactions on*

Artificial Intelligence, *4*(4), 820–839. https://doi.org/10.1109/TAI.2022.3225132

Tiwari, R., Rautela, S., Sharma, S., Pratap Choudhary, B., Tripathi, R., & Singh, P. (2023). Role of AI for Fraud Detection in Banks: A Bibliometric Analysis. *2023 International Conference on Advanced Computing & Communication Technologies (ICACCTech)*, 66–71. https://doi.org/10.1109/ICACCTech61146.2023.00020

Vakkuri, V., Kemell, K.-K., Kultanen, J., & Abrahamsson, P. (2020). The Current State of Industrial Practice in Artificial Intelligence Ethics. *IEEE Software*, *37*(4), 50–57. https://doi.org/10.1109/MS.2020.2985621

Venkatesh, V., Morris, M., Davis, G., & Davis, F. (2003). User Acceptance of Information Technology: Toward a Unified View. *MIS Quarterly*, *27*, 425–478. https://doi.org/10.2307/30036540

Venkatesh, V., Thong, J. Y. L., & Xu, X. (2012). Consumer Acceptance and Use of Information Technology: Extending the Unified Theory of Acceptance and Use of Technology. *MIS Quarterly*, *36*(1), 157–178. https://doi.org/10.2307/41410412

Vincent, J. (2019, November 11). *Apple's credit card is being investigated for discriminating against women*. The Verge. https://www.theverge.com/2019/11/11/20958953/apple-credit-card-gender-discrimination-algorithms-black-box-investigation

Waller, M., Rodrigues, O., Lee, M. S. A., & Cocarascu, O. (2025). Bias Mitigation Methods: Applicability, Legality, and Recommendations for Development. *J. Artif. Int. Res.*, *81*. https://doi.org/10.1613/jair.1.16759

Zotero. (2025). *Zotero | Your personal research assistant*. https://www.zotero.org/

APPENDIX 1

TABLE 6 The details of key ethical challenges in literature

Author, year	Data privacy & security	Bias & Discrimination	Transparency	Accountability	Fairness	Human displacement	Explainability	Empathy
(Boustani, 2022)	✓							✓
(Hentzen et al., 2022)	✓	✓	✓	✓		✓		
(Arora et al., 2023)	✓							
(Mohsen et al., 2024)	✓	✓		✓		✓	✓	
(Mogaji & Nguyen, 2022)	✓	✓	✓	✓				
(Eskandarany, 2024)	✓	✓	✓	✓	✓			
(Durongkadej et al., 2024)	✓	✓	✓					
(Baffour Gyau et al., 2024)		✓	✓	✓				
(Fundira et al., 2024)	✓	✓	✓		✓	✓		
(Manta et al., 2024)	✓	✓	✓		✓			
(Al-Baity, 2023)	✓	✓	✓	✓		✓		
(Sachdeva et al., 2024)		✓		✓	✓	✓		
(Sadok et al., 2023)	✓		✓	✓	✓			
(Maurya et al., 2024)	✓	✓	✓	✓				
(Kanjolia et al., 2024)	✓	✓	✓	✓				
(Mehndiratta et al., 2023)	✓							
(ALMahadin et al., 2023)	✓							
(Ramachandran et al., 2024)	✓	✓						
(Meena et al., 2024)	✓							
(Beckmann & Hark, 2024)	✓		✓	✓				
(Chernysh et al., 2024)	✓	✓						
(Gigante & Zago, 2023)	✓	✓		✓			✓	
(Manser Payne et al., 2021)	✓					✓		

Author, year	Data privacy & security	Bias & Discrimi-nation	Transpar-ency	Accounta-bility	Fair-ness	Human displace-ment	Explaina-bility	Empa-thy
(Barile et al., 2024)	✓	✓	✓			✓		
(Piotrowski & Orzeszko, 2023)	✓	✓	✓					
(Flavián et al., 2022)	✓	✓	✓					
(Pfoertsch & Sulaj, 2023)	✓	✓						✓
(Rohella et al., 2024)	✓							
(Kadam et al., 2024)	✓	✓			✓	✓		
(Jagadees-wari et al., 2024)	✓	✓	✓			✓		
(Akre et al., 2024)	✓	✓	✓	✓				
(Johora et al., 2024)	✓		✓					
(Tiwari et al., 2023)	✓	✓	✓	✓				
(Kö-nigstorfer & Thalmann, 2020)	✓	✓	✓	✓				
(Kumar et al., 2021)	✓	✓	✓	✓				
(Mathen & Paul, 2025)	✓	✓	✓	✓				
(Abakarim et al., 2018)	✓							
(Chitimira et al., 2024)	✓	✓	✓			✓		
(Balaji, 2024)	✓	✓	✓	✓				
(Lagasio et al., 2022)	✓	✓	✓	✓				
(Al Batayneh et al., 2021)					✓			
(Mishra, 2024)	✓	✓	✓	✓	✓			
(Iturbe et al., 2023)	✓					✓		
(Hanif et al., 2021)	✓	✓	✓	✓	✓			
(Adams & Hagras, 2020)				✓			✓	
(Adamyk et al., 2024)	✓	✓	✓	✓	✓		✓	
(Guler et al., 2024)	✓	✓	✓	✓			✓	
(Jáuregui-Velarde et al., 2024)	✓							
(Raman & Tiwari, 2024)			✓					
(Gaurav & Tiwari, 2023)			✓					
(Nagendiran et al., 2023)		✓						
(Abdul-Al et al., 2024)	✓							

Author, year	Data privacy & security	Bias & Discrimi-nation	Transpar-ency	Accounta-bility	Fair-ness	Human displace-ment	Explaina-bility	Empa-thy
(Sekar et al., 2024)	✓	✓	✓	✓				
(Milossi et al., 2021)	✓	✓	✓	✓		✓		
(Kamarud-din et al., 2023)	✓	✓	✓	✓	✓			
(Botunac et al., 2024)	✓							
(Tidjon & Khomh, 2023)	✓	✓	✓	✓	✓			
(Cath, 2018)	✓	✓	✓	✓		✓		
(Evangelista et al., 2023)	✓	✓	✓	✓	✓	✓		
(Piotrowski, 2023)	✓	✓	✓	✓				
(Nabbosa, 2020)	✓	✓			✓			
(Nguyen et al., 2024)	✓	✓	✓	✓	✓			
(Manoj Kumar et al., 2023)		✓		✓		✓		
(Chen, 2020)			✓	✓				
(Saleh, 2024)	✓	✓	✓	✓	✓			
(Andreasyan et al., 2024)	✓	✓	✓	✓				
(Ridzuan et al., 2024)	✓	✓	✓	✓	✓			
(Rawas, 2024)	✓	✓	✓	✓		✓		
(Divi et al., 2024)	✓	✓	✓	✓	✓			
(Aitken et al., 2021)	✓	✓	✓	✓				
Author, year	**Data privacy & security**	**Bias & Discrimi-nation**	**Transpar-ency**	**Accounta-bility**	**Fair-ness**	**Human displace-ment**	**Explaina-bility**	**Empa-thy**
Total #	**61**	**51**	**46**	**40**	**18**	**16**	**05**	**02**

APPENDIX 2

Categorization of the Ethical challenges of AI adoption according to my synthesis is detailed below:

TABLE 6 depicts Category 1: Ethical challenges of AI adoption in general perspective

A: AI adoption in banks (Implicit in ethical challenges)
B: AI adoption in specific areas in banking operations; customer service, mobile banking (General)
C: AI for fraud detection (General)
D: AI for credit scoring (General)
E: AI for money laundering detection (General)
F: AI for debt recovery (General)

TABLE 7 depicts Category 2: AI adoption in banking in technical perspective

A: AI adoption in banks for cybersecurity enforcement
B: Explainable AI (XAI) adoption in banking operations
C: Machine Learning (ML) adoption in banking operations
D: Biometric analysis with AI adoptability in banking

TABLE 8 depicts Category 3: AI adoption in banking in legal perspective

A: AI adoption in banking with relevance to GDPR
B: AI adoption in banking with relevance to other legal frameworks

TABLE 9 depicts Category 4: Ethical challenges of AI adoption in banking (explicit perspective)

A: Data privacy and security challenges
B: Accountability & liability
C: AI system development challenges
D: Other ethical challenges (accountability, liability, bias, transparency …etc.)

TABLE 7 Categorization: AI adoption in general perspective

Author(s), year	Ethical challenge(s)	Methodology	Category
[1] (Boustani, 2022)	AI vs. Human empathy, privacy & data usage	Quantitative	A: AI adoption in banks (Implicit)
[2] (Hentzen et al., 2022)	Bias in decision making, transparency, accountability, data privacy, inequality, human displacement	SLR	
[3] (Arora et al., 2023)	Data privacy & security, technology-induced stress	Qualitative-empirical and descriptive research	
[4] (Mohsen et al., 2024)	Data privacy & security, bias & accuracy, accountability & explainability, human displacement	Mixed methods (qualitative & quantitative)	
[5] (Mogaji & Nguyen, 2022)	Data privacy, bias, transparency, accountability	Qualitative	
[6] (Eskandarany, 2024)	Data privacy & security, algorithmic bias & fairness, transparency, accountability, ethical hacking & data manipulation	Qualitative (exploratory)	
[7] (Durongkadej et al., 2024)	Data privacy & security, bias & discrimination, transparency	Quantitative (event & case study)	
[8] (Baffour Gyau et al., 2024)	Bias & discrimination, transparency & accountability	Quantitative	
[9] (Fundira et al., 2024)	Data privacy & security, transparency, fairness, bias, lack of human touch	Qualitative (semi-structured interviews)	
[10] (Manta et al., 2024)	Data privacy & governance, algorithmic bias & model transparency, operational risks, digital inequality	Quantitative (bibliometric analysis)	
[11] (Al-Baity, 2023)	Data privacy & security, bias in decision-making, transparency & accountability, job displacement	SLR	
[12] * (Sachdeva et al., 2024)	Loss of decision-making authority, accountability & fairness, human displacement	Mixed methods (quantitative & qualitative)	
[13] * (Sadok et al., 2023)	Data privacy, bias, transparency, accountability	SLR	
[14] * (Maurya et al., 2024)	Data privacy, bias, transparency, accountability	SLR	
[15] * (Kanjolia et al., 2024)	Data privacy, bias, transparency, accountability	Qualitative (desktop literature search, deductive)	
[16] * (Mehndiratta et al., 2023)	Data privacy & security	SLR	
[17] * (ALMahadin et al., 2023)	Data privacy, customer trust	Quantitative	
[18] * (Ramachandran et al., 2024)	Bias, security issues	Mixed methods (qualitative & quantitative)	
[19] (Meena et al., 2024)	Security, discrimination	Mixed methods (qualitative & quantitative)	

Author(s), year	Ethical challenge(s)	Methodology	Category
[20] (Beckmann & Hark, 2024)	Data privacy, customer trust, transparency, accountability	Quantitative	
[21] (Chernysh et al., 2024)	Data privacy & security, algorithmic bias, opacity of AI models	Quantitative-Empirical	
[22] (Gigante & Zago, 2023)	Data privacy & security, bias, accountability, explainability, labour market inequality	Mixed method (qualitative & quantitative)	B: AI adoption in specific areas in banking operations; customer service, mobile banking (General)
[23] (Manser Payne et al., 2021)	Data privacy & security, emotional detachment, loss of human interaction, trust issues	Quantitative	
[24] (Barile et al., 2024)	Data privacy & security, bias, transparency, lack of human interaction, service inequality	Qualitative (exploratory)	
[25] (Piotrowski & Orzeszko, 2023)	Data privacy, business ethics of banks (trust), transparency, bias (moral judgements)	Quantitative survey	
[26] (Flavián et al., 2022)	Data privacy, transparency, bias	Quantitative	
[27] * (Pfoertsch & Sulaj, 2023)	Data privacy, bias, empathy	Quantitative (empirical)	
[28] * (Rohella et al., 2024)	Data privacy & security	Mixed methods (qualitative & quantitative)	
[29] * (Kadam et al., 2024)	Data privacy & security, bias & fairness, cultural resistance (lack of human intervention)	Qualitative	
[30] * (Jagadeeswari et al., 2024)	Data privacy, bias, transparency, human displacement	Quantitative	
[31] * (Akre et al., 2024)	Data privacy, bias, accountability, transparency	Quantitative	
[32] * (Johora et al., 2024)	Privacy, transparency (focus on fraud detection using ML	Qualitative	C: fraud detection
[33] * (Tiwari et al., 2023)	Data privacy, bias, accountability, transparency	Quantitative (bibliometric analysis)	
[34] (Königstorfer & Thalmann, 2020)	Data privacy, algorithmic bias, accountability, transparency	SLR	D: AI for credit scoring (General)
[35] (Kumar et al., 2021)	Data privacy, algorithmic bias & discrimination, transparency, misuse & manipulation	SLR	
[36] (Mathen & Paul, 2025)	Data privacy, algorithmic bias & discrimination, transparency (opacity of models), Incompatibility	SLR	
[37] * (Abakarim et al., 2018)	Ethical use of data	Quantitative	

Author(s), year	Ethical challenge(s)	Methodology	Category
[38] (Chitimira et al., 2024)	Data privacy & cybersecurity, bias & false positives, transparency, human disempowerment	Qualitative	E: AI for money laundering
[39] * (Balaji, 2024)	Data privacy, algorithmic bias & discrimination, transparency, accountability	Quantitative (bibliometric analysis)	
[40] (Lagasio et al., 2022)	Data privacy, algorithmic bias, transparency, accountability	Quantitative (ML algorithms)	F: for Debt recovery
[41] * (Dogra & Kumar, 2024)	Data privacy, bias, transparency, accountability & fairness, regulatory compliance	Qualitative (SLR & case studies)	

Category 2: AI adoption in banking in technical perspective

A: AI adoption in banks for cybersecurity enforcement

B: Explainable AI (XAI) adoption in banking operations

C: Machine Learning (ML) adoption in banking operations

D: Biometric analysis with AI adoptability in banking

TABLE 8 Categorization: AI adoption in banking in technical perspective

Author(s), year	Ethical challenge(s)	Methodology	Category
[42] (Al Batayneh et al., 2021)	Work ethics & values, trust, compliance issues	Quantitative	A: cybersecurity enforcement
[43] (Mishra, 2024)	Data privacy & security, bias, transparency, fairness	Mixed methods (qualitative & quantitative)	
[44] * (Iturbe et al., 2023)	Data privacy & security, trust, compliance with work ethics & fundamental rights	Qualitative	
[45] * (Hanif et al., 2021)	Data privacy, bias & fairness, accountability, responsibility, trust	Qualitative (survey)	B: Explainable AI (XAI) adoption
[46] * (Adams & Hagras, 2020)	Loss of confidence & trust, accountability & explainability	Quantitative	
[47] * (Adamyk et al., 2024)	Data privacy & security, bias, fairness, transparency, explainability	Qualitative (semi-structured interviews)	
[48] (Guler et al., 2024)	Data privacy, algorithmic bias, transparency & explainability	Quantitative (Computational Literature Review)	C: Machine Learning (ML) adoption
[49] (Jáuregui-Velarde et al., 2024)	Data privacy, malicious manipulation, trust issues	Quantitative (Systematic Review of Literature)	
[50] * (Bhatt & Meniya, 2022)	None	Quantitative (ML algorithms)	
[51] * (Raman & Tiwari, 2024)	Transparency	Quantitative (ML algorithms)	

Author(s), year	Ethical challenge(s)	Methodology	
[52] * (Gaurav & Tiwari, 2023)	Transparency, Trust	Quantitative	
[53] * (Nagendiran et al., 2023)	Bias	Quantitative (ML algorithms)	
[54] (Abdul-Al et al., 2024)	Data privacy	Quantitative	D:Biometric analysis
[55] * (Sekar et al., 2024)	Data privacy, algorithmic bias & discrimination, transparency	Qualitative-SLR	

Category 3: AI adoption in banking (Legal Perspective)

A: AI adoption in banking with relevance to GDPR

B: AI adoption in banking with relevance to other legal frameworks

TABLE 9 Categorization: AI adoption in banking in legal perspective

Author(s), year	Ethical challenge(s)	Methodology	Category
[56] (Milossi et al., 2021)	Data privacy, bias & discrimination, transparency & accountability, autonomy	Qualitative	A: GDPR framework
[57] * (Kamaruddin et al., 2023)	Data protection & privacy, bias & discrimination, algorithmic accountability, surveillance, compliance with regulations	Qualitative	
[58] * (Botunac et al., 2024)	Data privacy & trust	Qualitative	
[59] (Tidjon & Khomh, 2023)	Data privacy & protection, bias & fairness, transparency & accountability, ethical governance	Qualitative	B: Other legal frameworks
[60] (Cath, 2018)	Data privacy, accountability & transparency, human displacement	Qualitative	
[61] * (Evangelista et al., 2023)	Data privacy, trust, bias & discrimination, accountability, job displacement	Qualitative	

Category 4: Ethical challenges of AI adoption in banking (explicit perspective)

A: Data privacy and security challenges
B: Accountability & liability
C: AI system development challenges
D: Other ethical challenges (fairness, explainability, trust, empathy, human displacement concerns and …etc.)

TABLE 10 Categorization: AI adoption in banking in legal perspective

Author(s), year	Ethical challenge(s)	Methodology	Category
[62] (Piotrowski, 2023)	Data privacy, bias & accountability, transparency, ethical perception	Quantitative	A: Data privacy & Security
[63] * (Nabbosa, 2020)	Data privacy, dignity & discrimination, trust & reputation, compliance concerns	Qualitative	
[64] * (Nguyen et al., 2024)	Data privacy & security, bias & fairness, transparency & accountability	Qualitative	
[65] * (Manoj Kumar et al., 2023)	Algorithmic disclosure, appeal mechanism, human oversight, ethical guidelines-accountability & liability	Qualitative	B:Accountability & liability
[66] * (Chen, 2020)	Discrimination & bias, transparency & trust, monitoring & regulation, safeguard mechanisms	Qualitative	C: AI system development
[67] * (Saleh, 2024)	Data privacy, bias & fairness, accountability, transparency, regulatory compliance	Qualitative	
[68] * (Andreasyan et al., 2024)	Data privacy, transparency, security, fairness, responsibility	Quantitative	D: Other ethical challenges (explicit)
[69] (Ridzuan et al., 2024)	Data privacy & security, bias & fairness, accountability & transparency	Qualitative- SLR	
[70] (Rawas, 2024)	Bias & fairness, transparency & accountability, privacy & data security, human displacement	Quantitative-empirical	
[71] (Divi et al., 2024)	Bias, misinformation, privacy, accountability, transparency, discrimination	Qualitative	
[72] (Aitken et al., 2021)	Data privacy, bias & discrimination, transparency & accountability	Qualitative	